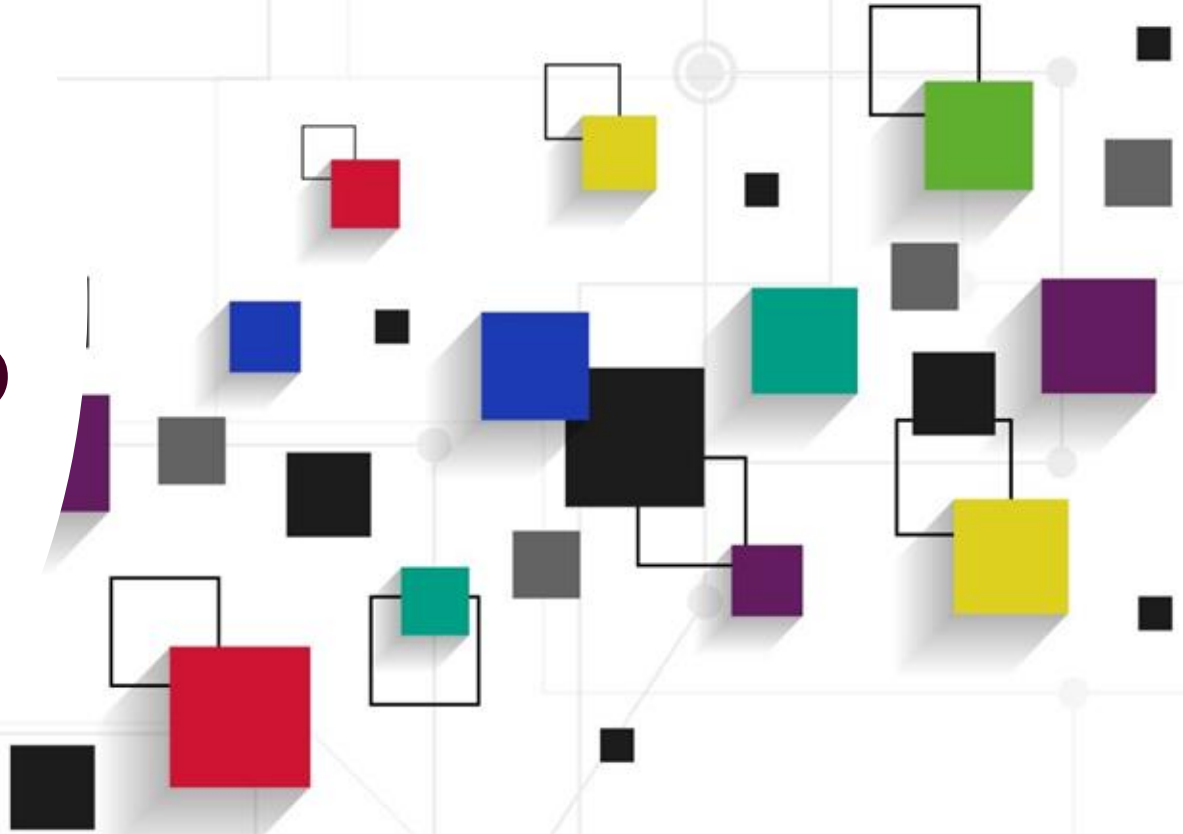
Pengenalan Hadoop

Pertemuan 6 - Teknologi Perekayasaan Data

Politeknik Statistika STIS

Siti Mariyah, Ph.D



Agenda Pertemuan

1. Big Data dengan Hadoop
2. Hadoop Architecture
3. Bagaimana Hadoop Bekerja
4. Virtualization, Containerization



In a farm far away...

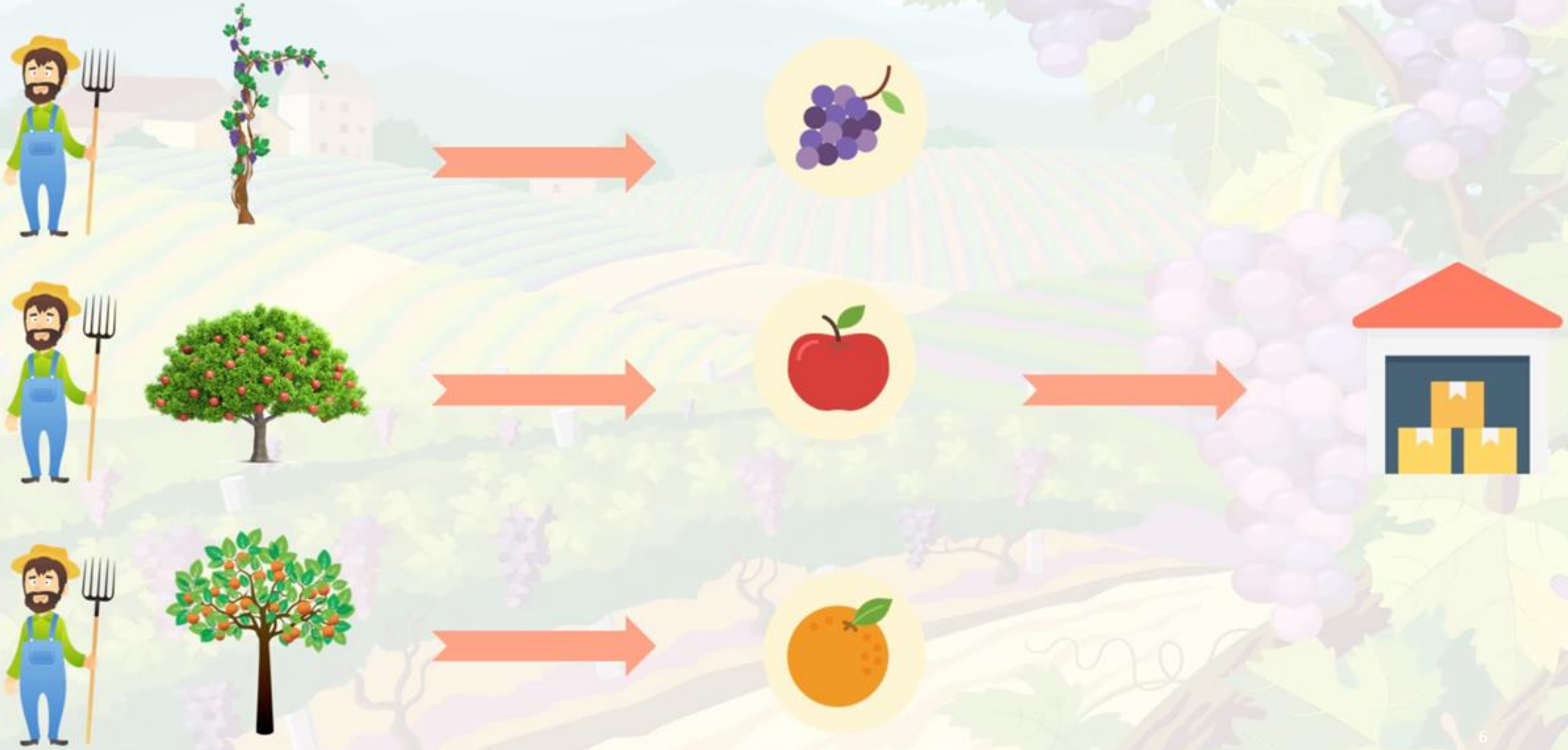
Jack harvests grapes and then sells it in the nearby town



After harvesting, he then stores the produce in a storage room



Soon there was a high demand for other fruits. So, he started harvesting apples and oranges as well



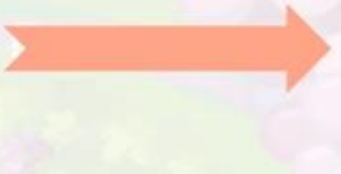
He then realizes that it is time consuming and difficult to harvest all the fruits by himself







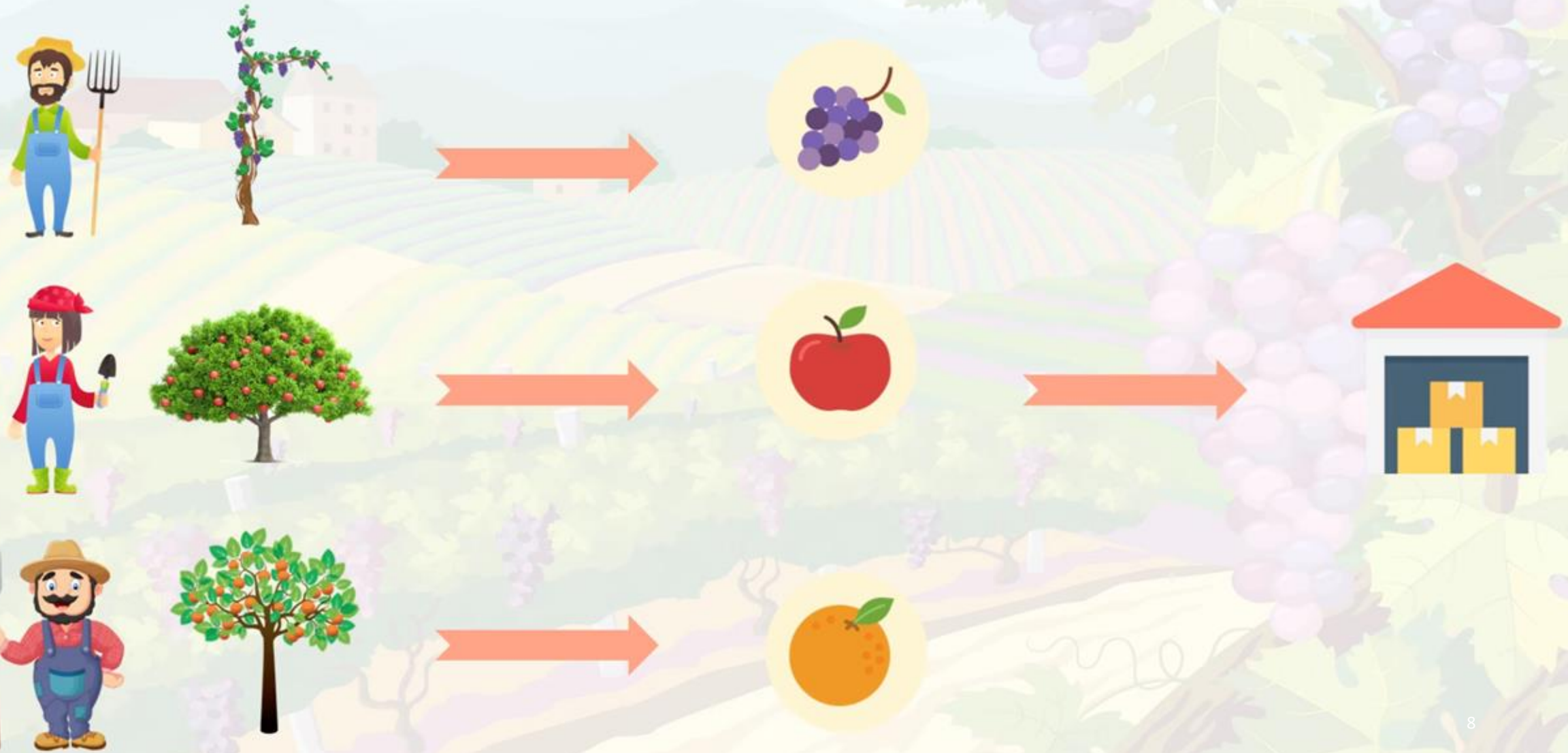









So, he hires 2 more people to work with him.
With this, harvesting is done simultaneously



Jack now decides to distribute the storage area and give each one of them a separate storage space















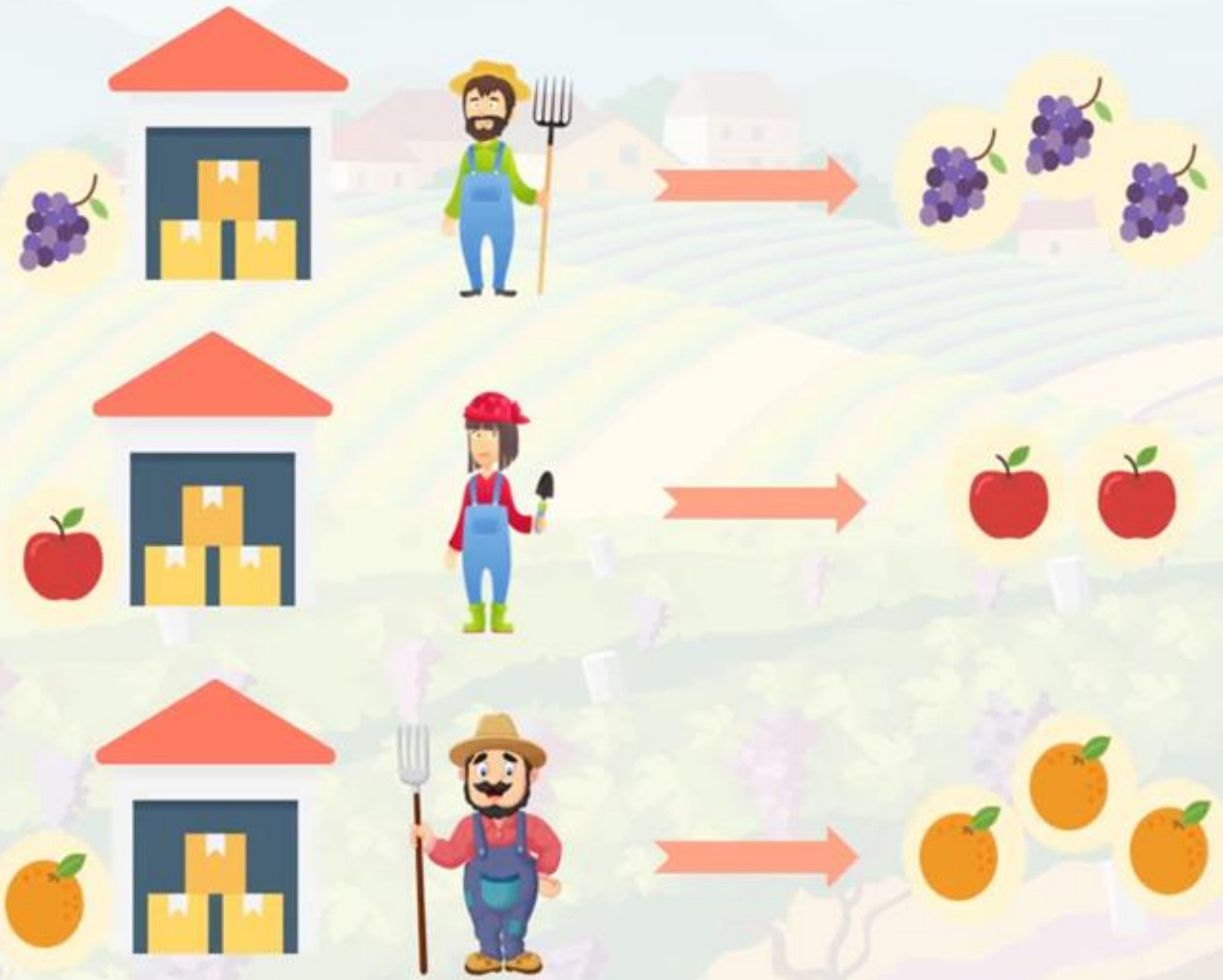








To complete the order on time, all of them work parallelly with their own storage space



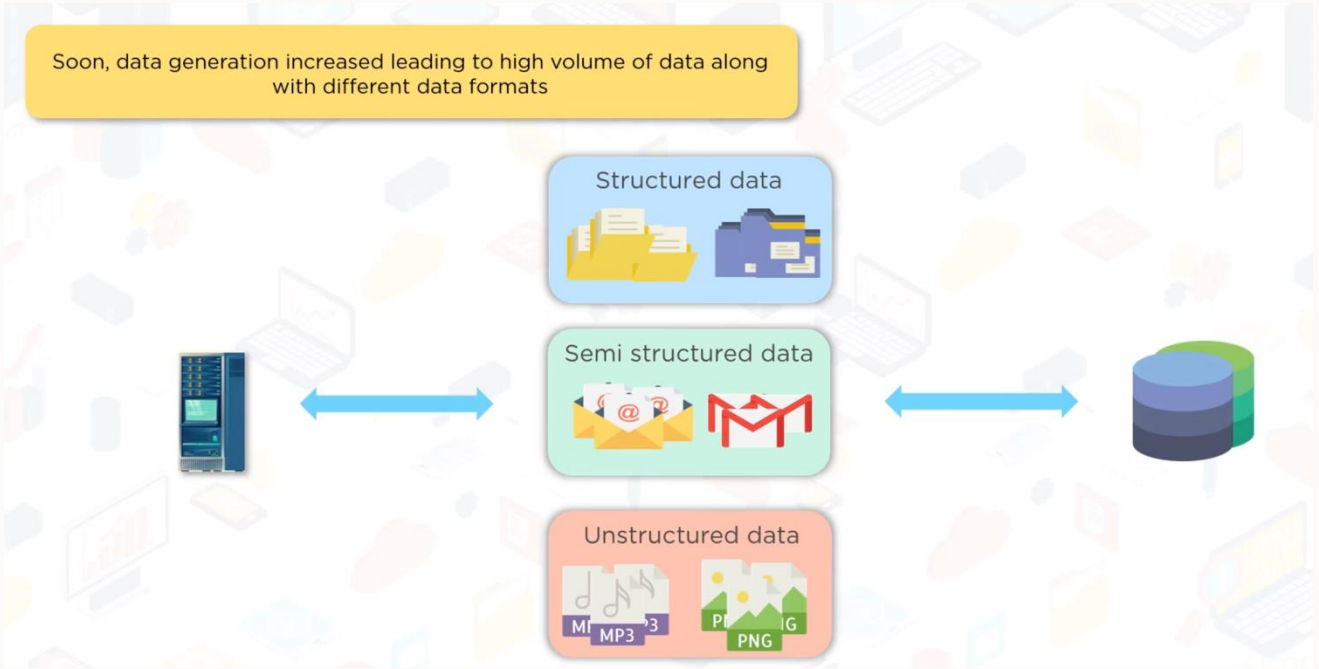
Hello, I want a fruit basket of 3 grapes, 2 apples and 3 oranges



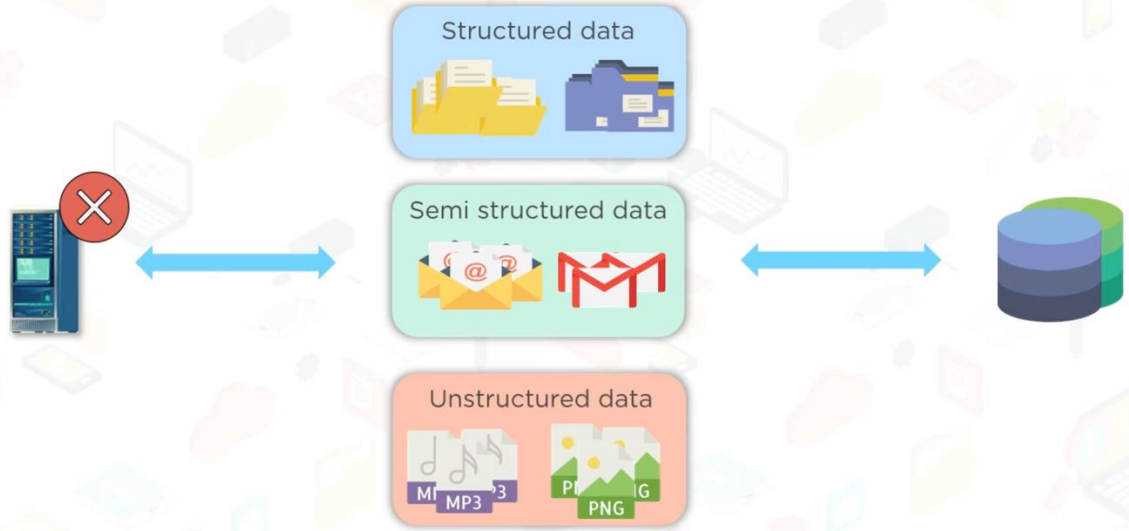
All of them are happy and they are prepared for an increase in demand in the future



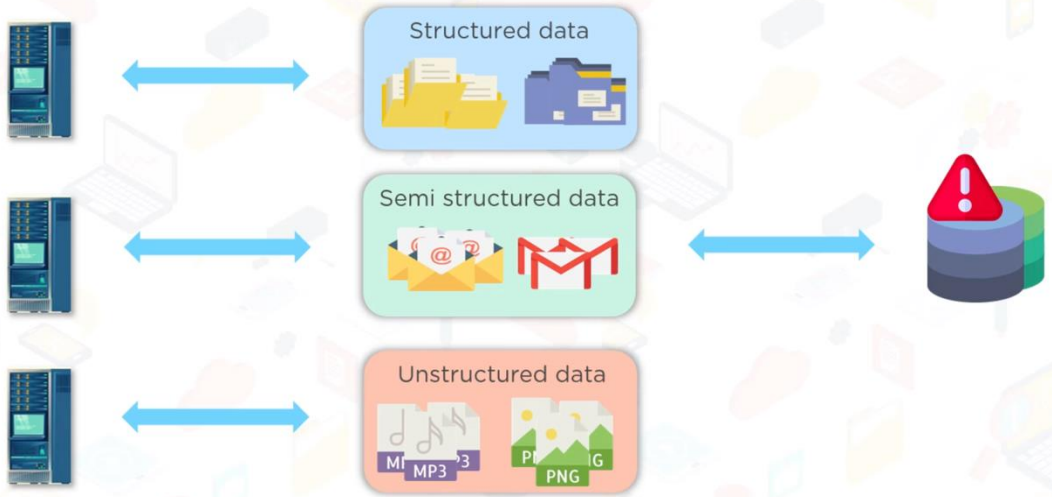

Earlier with limited data, only one processor and one storage unit was needed
Structured data


Soon, data generation increased leading to high volume of data along with different data formats
Structured data
Semi structured data
Unstructured data
MP3
PNG

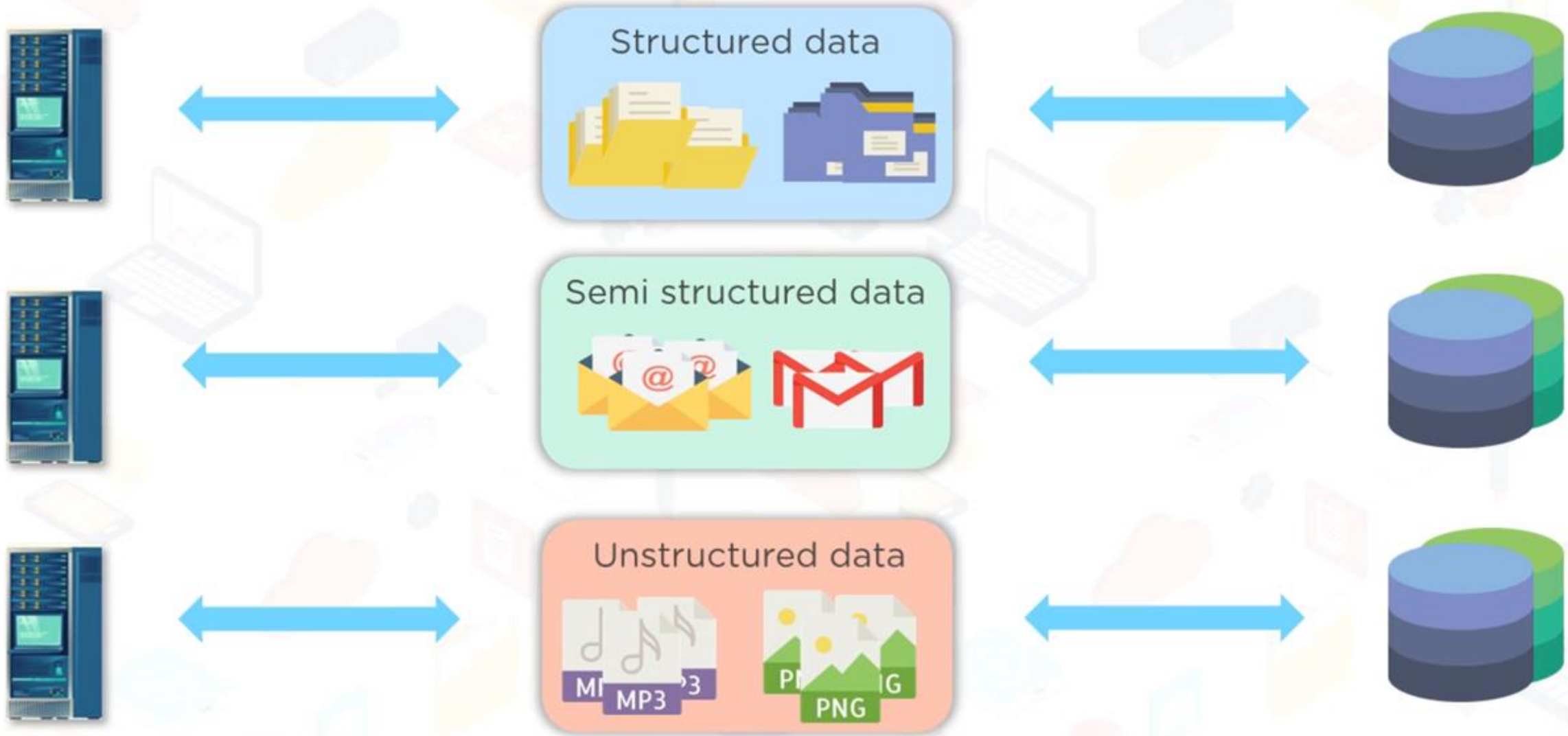
A single processor was not enough to process such high volume of different kinds of data as it was very time consuming



The single storage unit became the bottleneck due to which network overhead was generated



The solution was to use distributed storage for each processor. This enabled easy access to store and access data



What is Hadoop?

- **Hadoop** is an open-source software framework that is used for storing and processing large amounts of data in a distributed computing environment. It is designed to handle big data and is based on the MapReduce programming model, which allows for the parallel processing of large datasets.

Kelebihan
✓ **Scalability**: data are divided multiple machines in a cluster which is processed parallelly
✓ **Cost-effective**: open-source (free to use), commodity hardware (inexpensive, widely available and more or less interchangeable)
✓ **Fault-tolerance**: data are replicated on various nodes, if the machine faces a technical issue, data can also be read from other nodes
✓ **Flexibility**: it can deal with any kind of dataset
✓ **Open-source**
✓ **Large community**
✓ **Integration**: multiple, disparate internal and external sources and putting it in a single location

Kekurangan ?

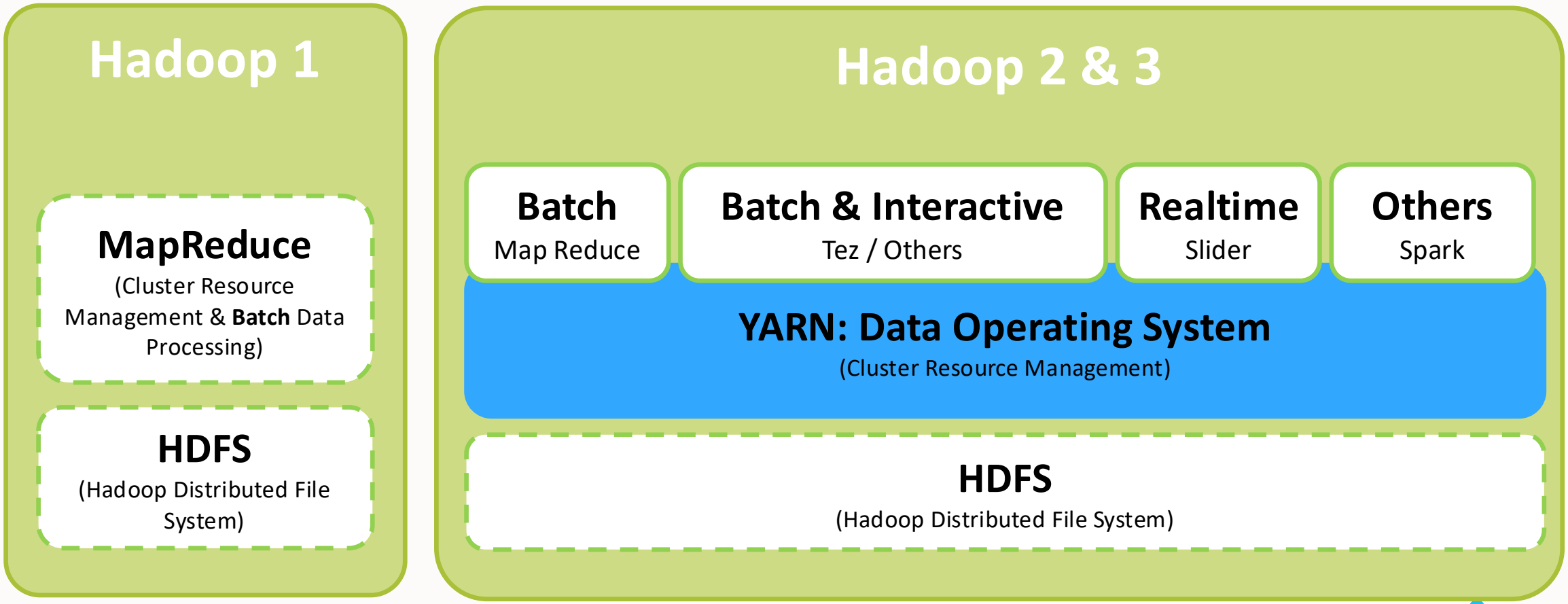
Komponen (Utama) Hadoop



Storage unit of Hadoop

Processing unit of Hadoop

Evolusi Arsitektur Hadoop



Komponen Utama Hadoop (Evolusi)

- **Hadoop Common**: Berisikan Libraries Java dan Utilities yang dibutuhkan modul lainnya
- **HDFS (Hadoop Distributed File System)**: Menyediakan akses ke data aplikasi *(distributed storage)*
- **Yarn (Yet Another Resource Negotiator)**: Mengelola Cluster Resources dan Job Scheduling
- **MapReduce**: Pemrosesan dataset secara parallel *(distributed processing)*

How Hadoop works?



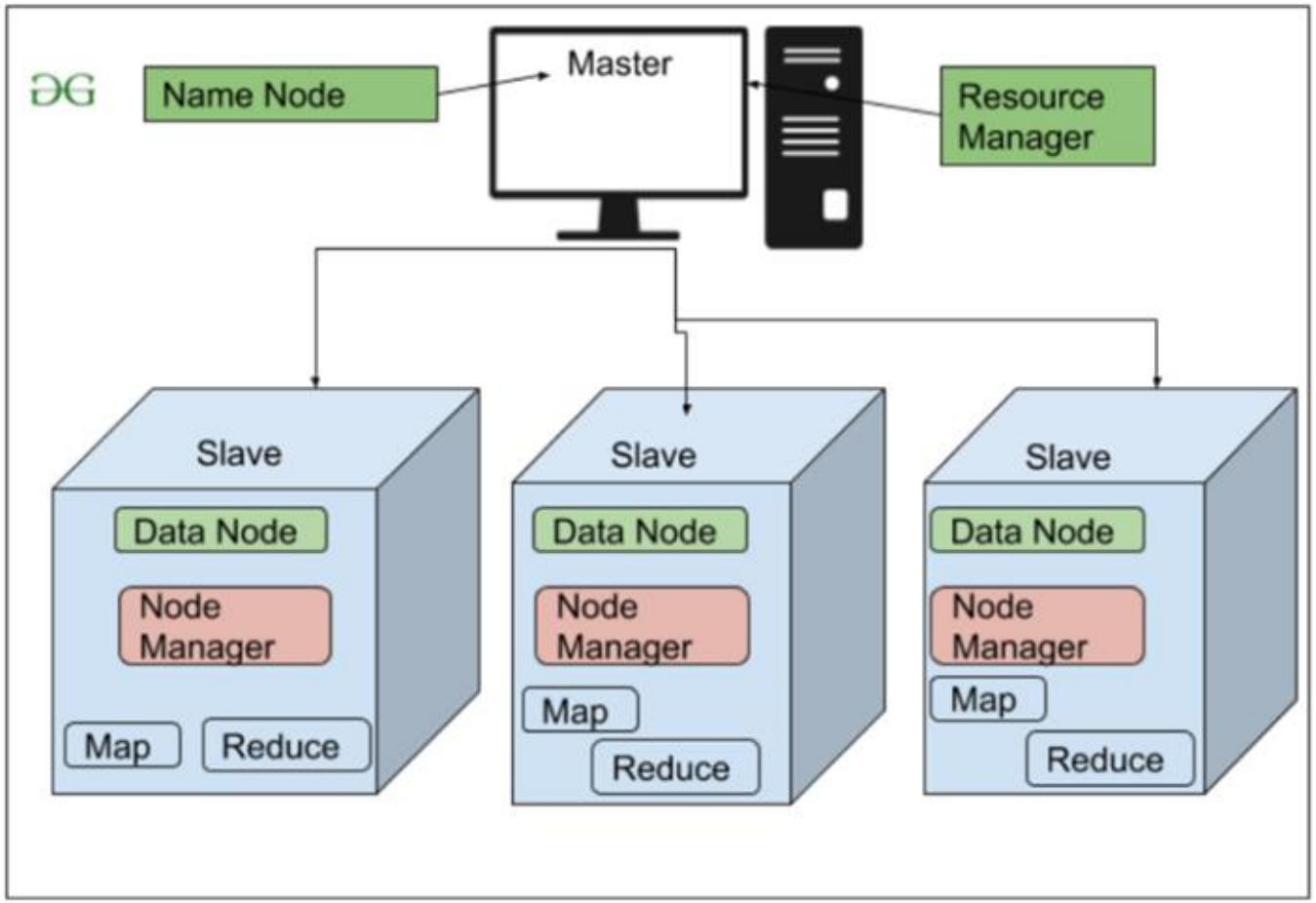
~ > 1 Tahun

2 Bulan ~

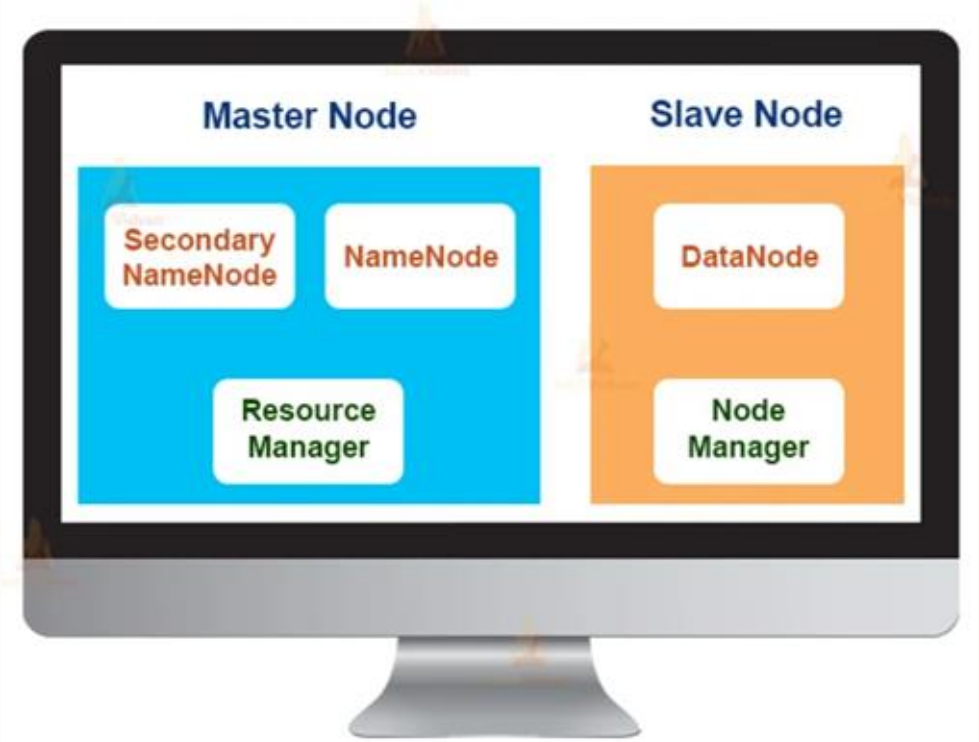


Gambaran Cluster Hadoop (**Multi Node Hadoop Cluster**)

Master-Slave architecture
- NameNode (Master)
- DataNode (Slave/Worker)



Bagaimana dengan **Single Node Hadoop Cluster**?

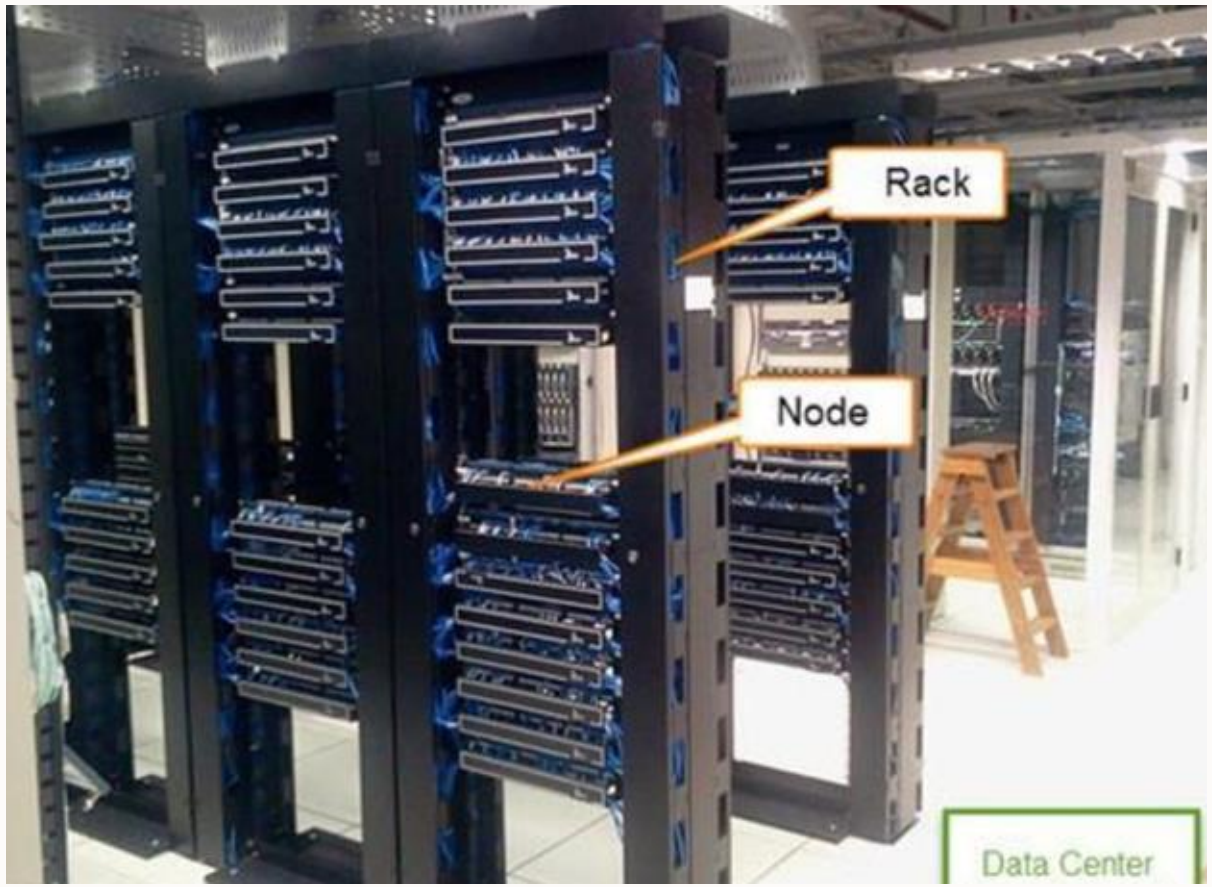


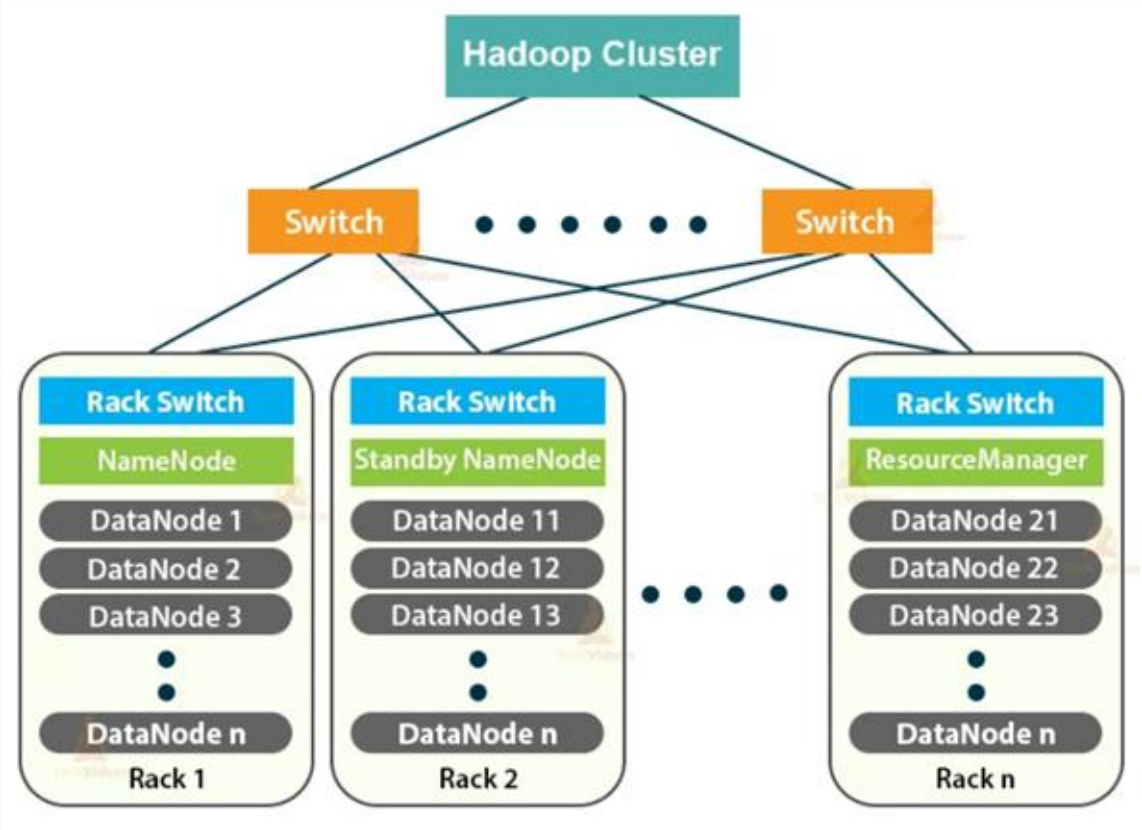
Single Node Hadoop Cluster diterapkan pada **satu mesin**. Semua *daemon (services)* seperti NameNode, DataNode, ResourceManager, NodeManager berjalan di mesin/host yang sama.

Perbedaannya:

- **Single node cluster**: hanya 1 DataNode yang *running*
- **Multi node cluster**: lebih dari 1 DataNode yang *running* dan setiap DataNode berjalan di mesin berbeda

Gambaran Cluster Hadoop

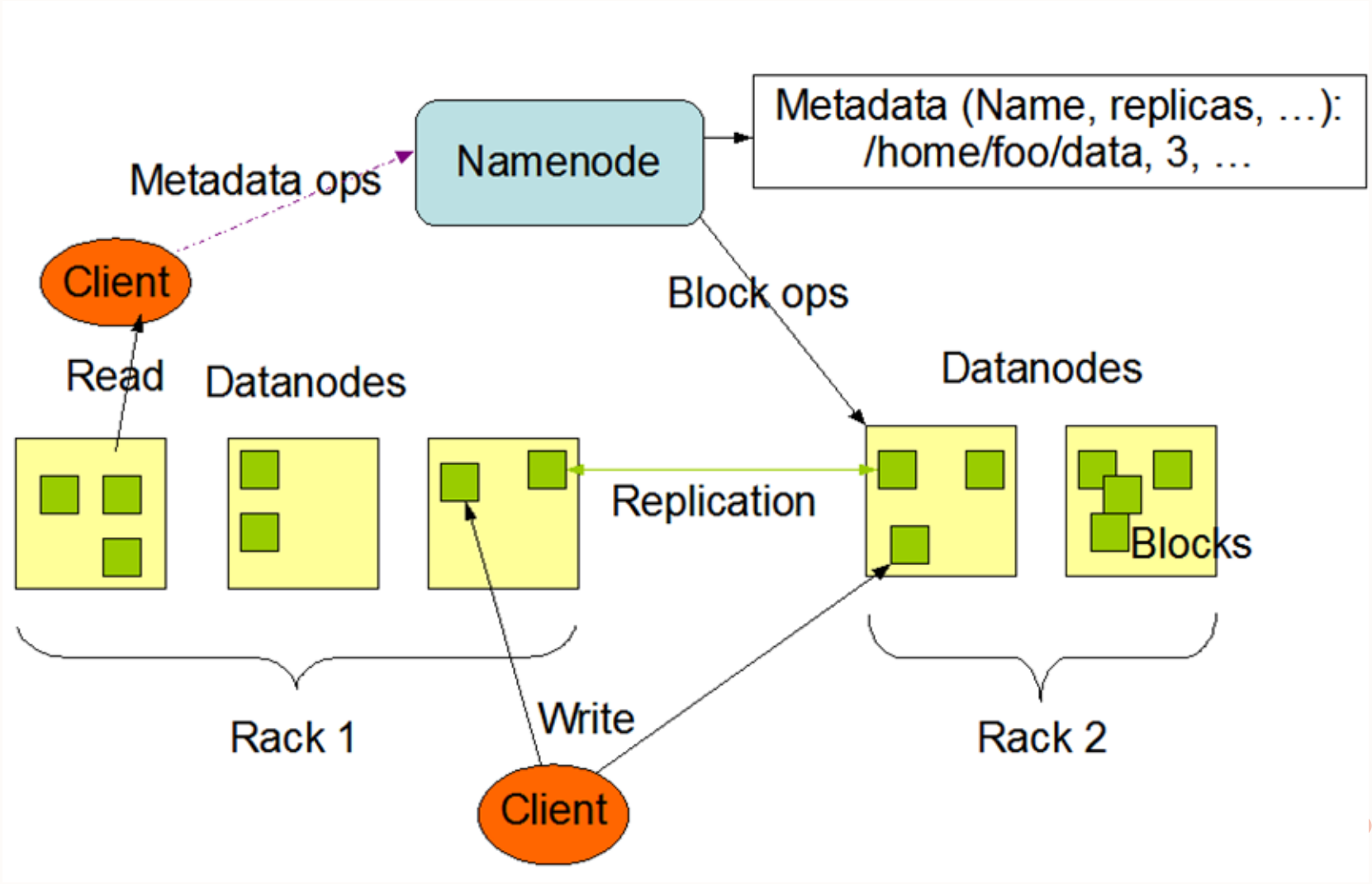




How Hadoop works?

Cluster	• Sekumpulan komputer yang terdiri DataNodes dan NameNode
NameNode	• Komputer yang bertugas sebagai monitor dan supervisor operasi-operasi yang dilakukan DataNodes, menyimpan metadata (data about data)
DataNode	• Komputer yang bertugas menyimpan file data dan melakukan pemrosesan
Mapper	• Bertugas dalam melakukan pencarian blok data yang tersebar di DataNodes dan melakukan pemrosesan di mana blok data tersimpan
Reducer	• Bertugas dalam mengambil dan mengompilasi hasil pemrosesan oleh Mapper
Block	• Satuan terkecil dari file yang dipecah-pecah

How Hadoop works?



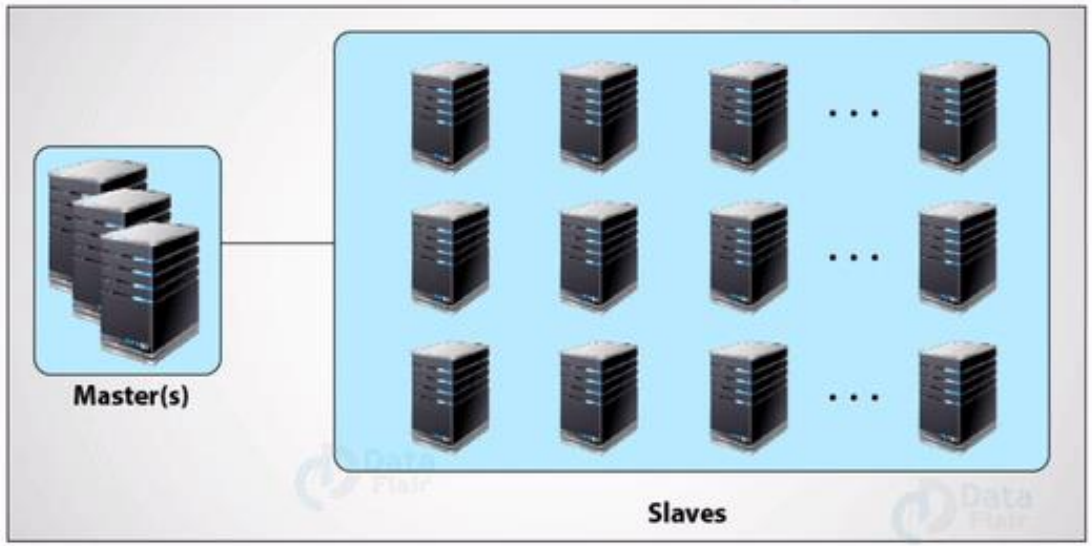
How Hadoop works?

1. Ketika suatu aplikasi **client** akan ***read/write*** data di Hadoop, mereka akan berinteraksi dengan **NameNode**. NameNode pertama-tama akan memeriksa hak akses client, dan apakah client memiliki permission. Jika ya, NameNode menanggapi permintaan dari client dengan menyediakan daftar *address* dari DataNodes dimana client dapat *read* atau *write* data. Setelahnya, aplikasi client dapat berinteraksi langsung ke DataNode.
2. **DataNodes** selalu terhubung ke NameNode. Mereka mengirimkan sinyal, yang dikenal sebagai *heartbeats*, ke NameNode. Heartbeats ini menunjukkan status DataNode. DataNode akan terus mengecek apakah ada permintaan dari NameNode untuk mengakses data.
3. **Misalkan client akan menambahkan data**, client dapat secara langsung *write* data pada DataNodes yang telah disediakan alamatnya oleh NameNode. File data client dalam HDFS dibagi menjadi beberapa blok. Ukuran blok adalah 128 MB secara default (bisa diganti). DataNode menyimpan blok file. Untuk memberikan **fault-tolerance**, HDFS membuat replika blok tergantung pada faktor replikasi, secara default adalah 3. Status apakah sudah selesai akan dikirimkan ke NameNode.

How Hadoop works?

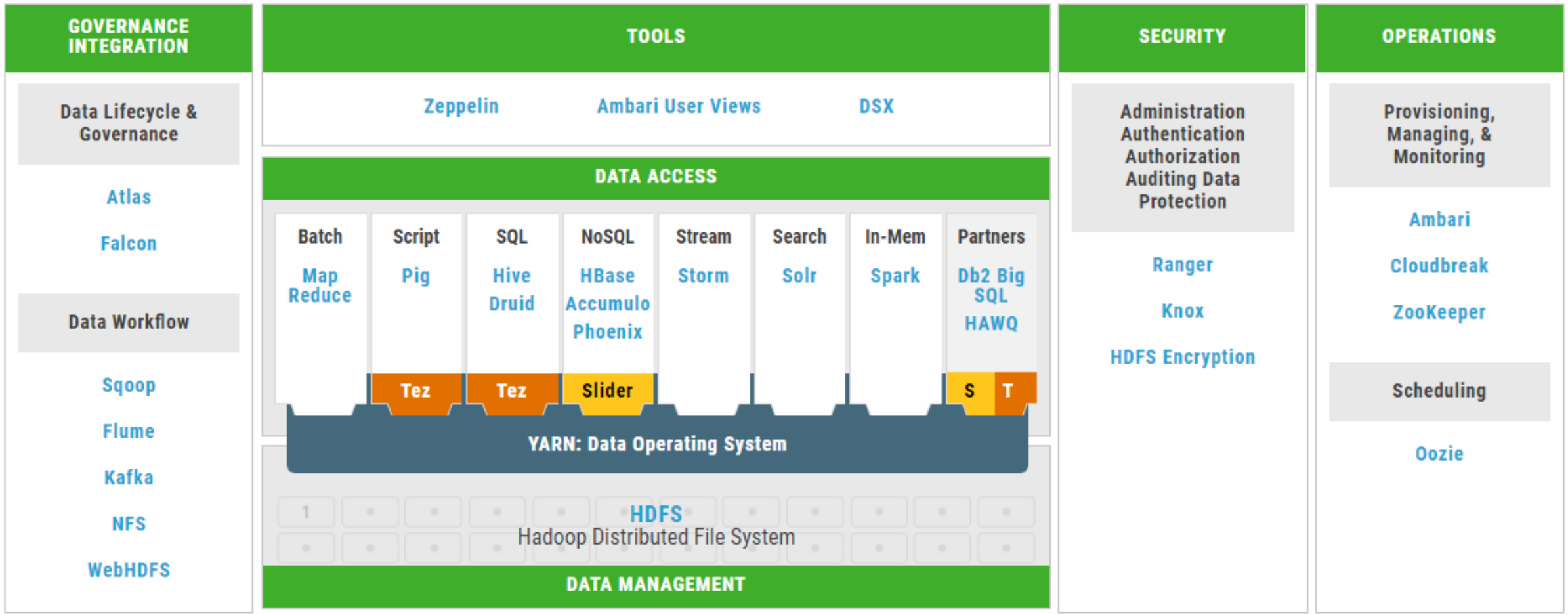
4. **Misalkan client akan membaca data**, client juga dapat secara langsung *read* data pada DataNodes yang telah disediakan alamatnya oleh NameNode. Client akan mengirim permintaan ke DataNodes terdekat. Data akan mengalir langsung dari DataNode ke client. Selama proses read, jika ada DataNode yang mati, NameNode memberikan alamat DataNode lain yang berisi replika blok dimana client dapat membaca datanya tanpa downtime. Status apakah sudah selesai akan dikirimkan ke NameNode.

Data Storage in HDFS



Hadoop Cluster

Arsitektur Hortonworks Data Platform (HDP)



Komponen Hadoop

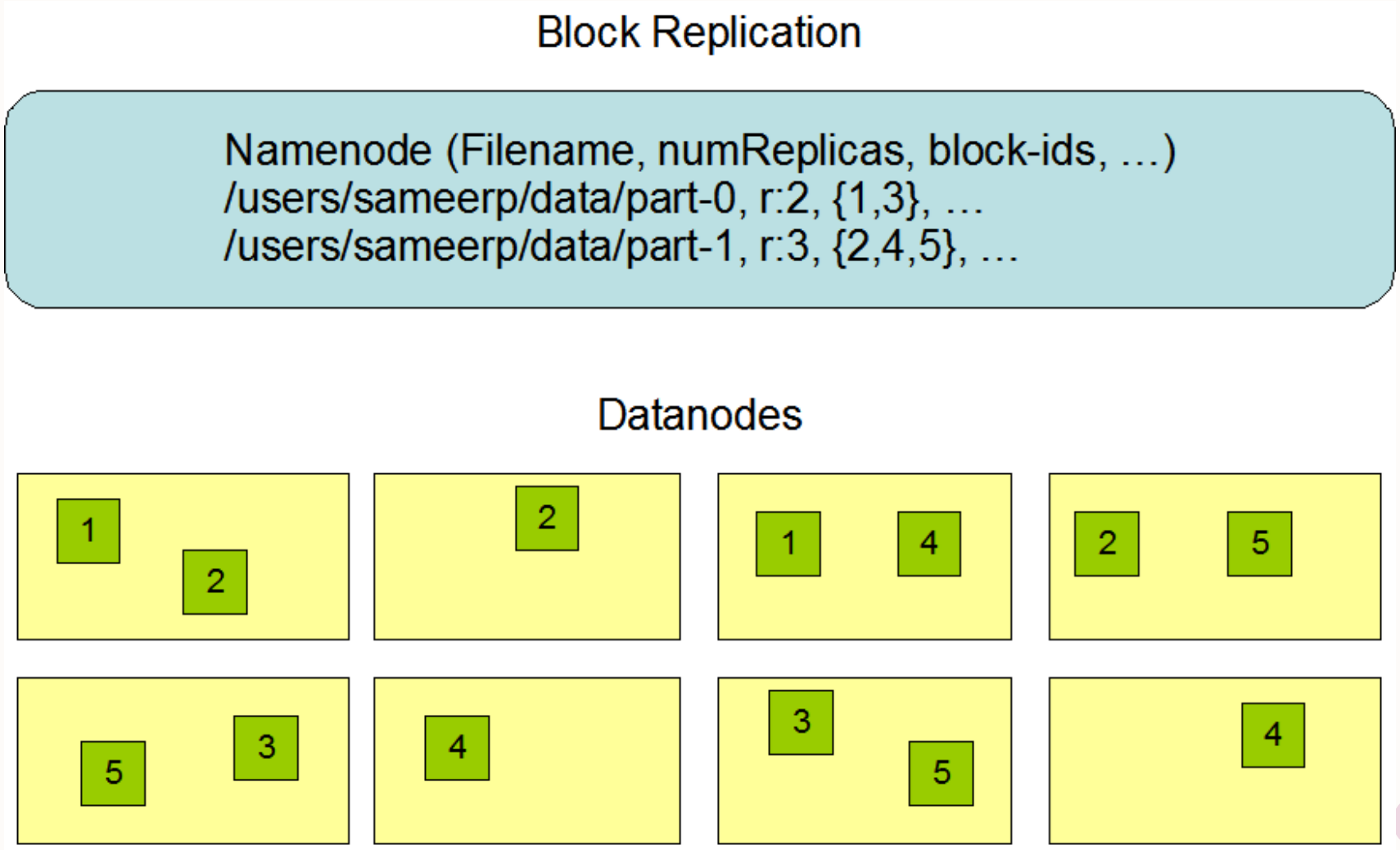
Table 1-1 Related Hadoop Projects

Project Name	*Description*
Ambari	An integrated set of Hadoop administration tools for installing, monitoring, and maintaining a Hadoop cluster. Also included are tools to add or remove slave nodes.
Avro	A framework for the efficient *serialization* (a kind of transformation) of data into a compact binary format
Flume	A data flow service for the movement of large volumes of log data into Hadoop
HBase	A distributed columnar database that uses HDFS for its underlying storage. With HBase, you can store data in extremely large tables with variable column structures
HCatalog	A service for providing a relational view of data stored in Hadoop, including a standard approach for tabular data

Table 1-1 *(continued)*

Project Name	*Description*
Hive	A distributed data warehouse for data that is stored in HDFS; also provides a query language that's based on SQL (HiveQL)
Hue	A Hadoop administration interface with handy GUI tools for browsing files, issuing Hive and Pig queries, and developing Oozie workflows
Mahout	A library of machine learning statistical algorithms that were implemented in MapReduce and can run natively on Hadoop
Oozie	A workflow management tool that can handle the scheduling and chaining together of Hadoop applications
Pig	A platform for the analysis of very large data sets that runs on HDFS and with an infrastructure layer consisting of a compiler that produces sequences of MapReduce programs and a language layer consisting of the query language named Pig Latin
Sqoop	A tool for efficiently moving large amounts of data between relational databases and HDFS
ZooKeeper	A simple interface to the centralized coordination of services (such as naming, configuration, and synchronization) used by distributed applications

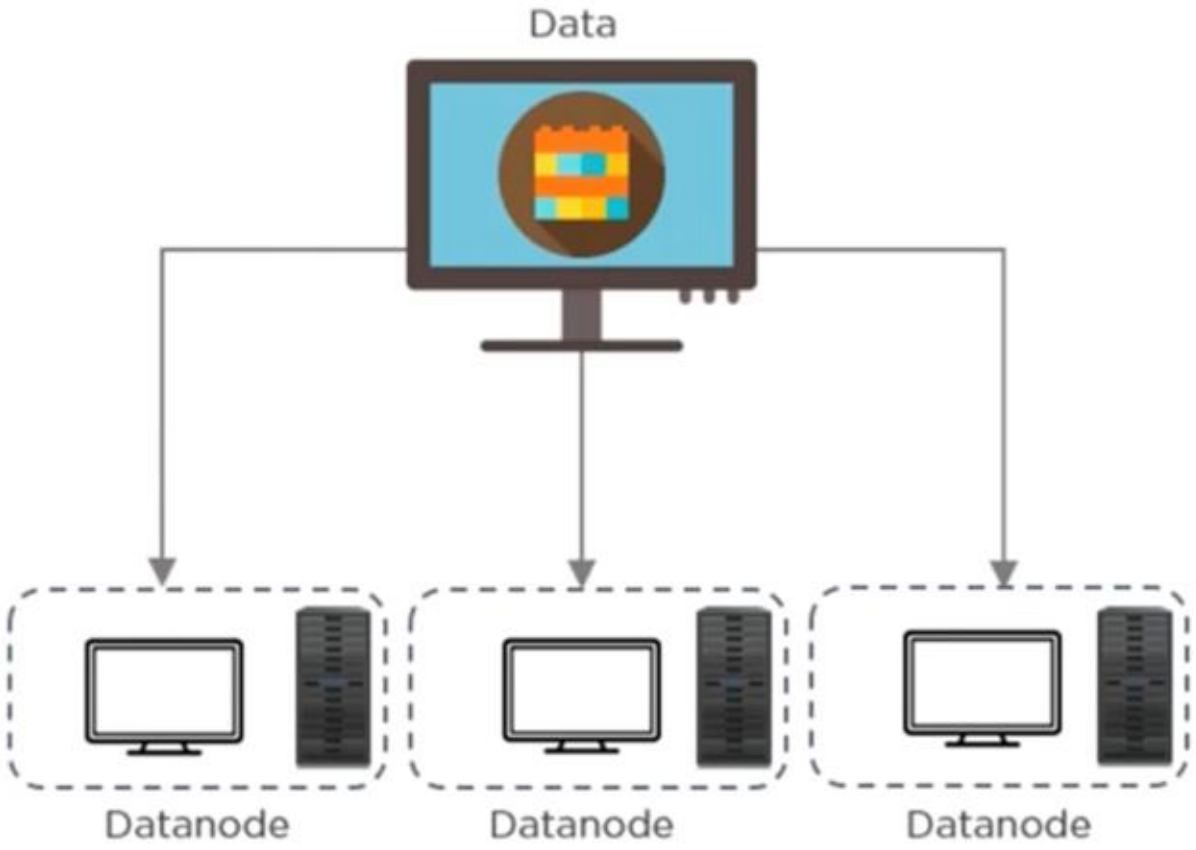
Komponen Hadoop - **HDFS**



Apa itu HDFS?

- **Sistem File Terdistribusi Hadoop (Hadoop Distributed File System - HDFS)** adalah sistem file terdistribusi yang dirancang untuk berjalan pada *perangkat keras komoditas*.
- HDFS memiliki banyak kesamaan dengan sistem file terdistribusi yang ada. Namun, perbedaan dengan sistem file terdistribusi lainnya sangat signifikan. HDFS sangat toleran terhadap kesalahan dan dirancang untuk digunakan pada perangkat keras berbiaya rendah. HDFS menyediakan akses *throughput* tinggi ke data aplikasi dan cocok untuk aplikasi yang memiliki dataset besar. HDFS mengaktifkan akses streaming ke sistem file data.
- HDFS awalnya dibangun sebagai infrastruktur untuk proyek mesin pencari web Apache Nutch. HDFS adalah bagian dari proyek Apache Hadoop Core.

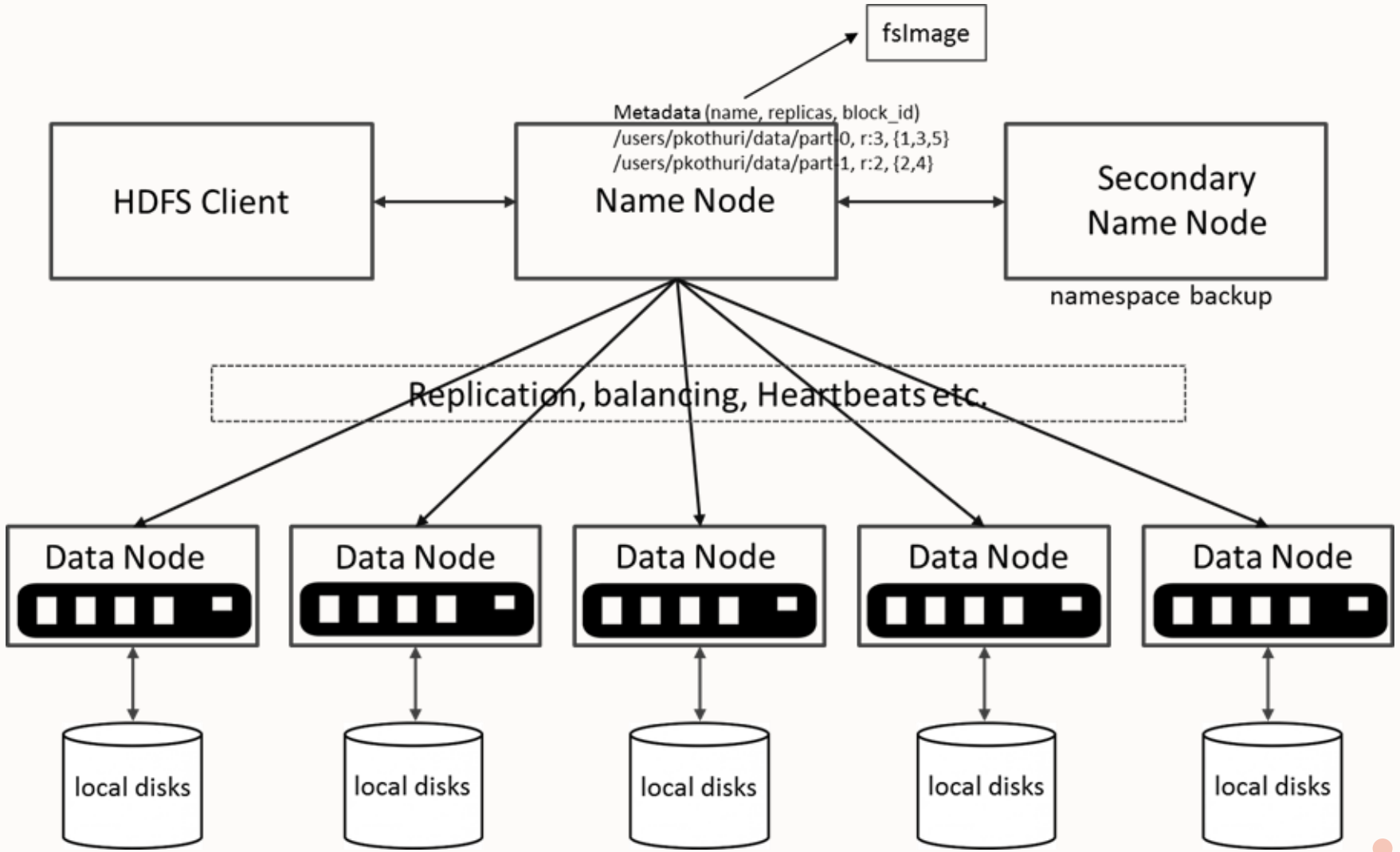
Hadoop Distributed File System (HDFS) is known for its distributed storage method. It distributes the data amongst many computers. In addition to this, replication of data is also done to avoid loss of data



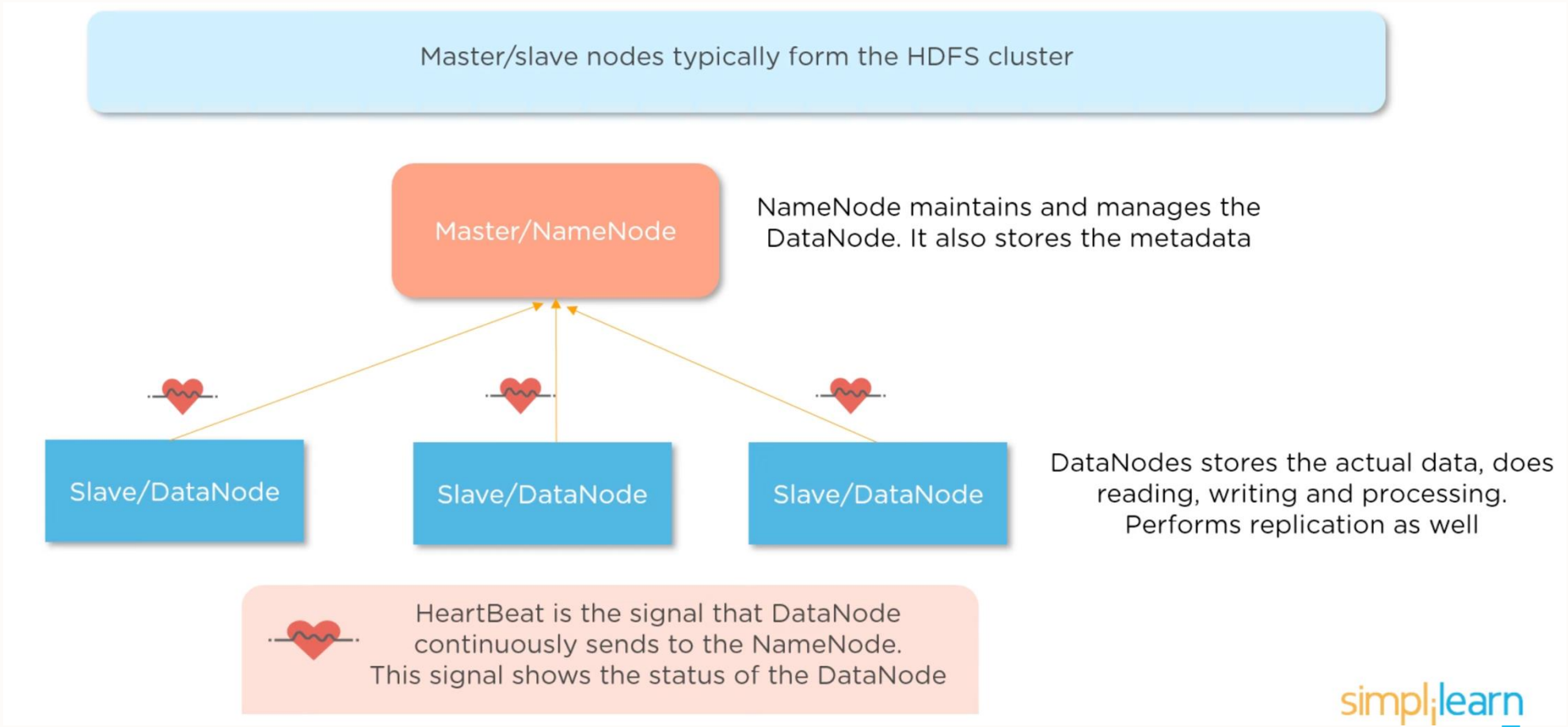
Each block of data is stored on multiple systems and by default has 128 MB of data

simplilearn

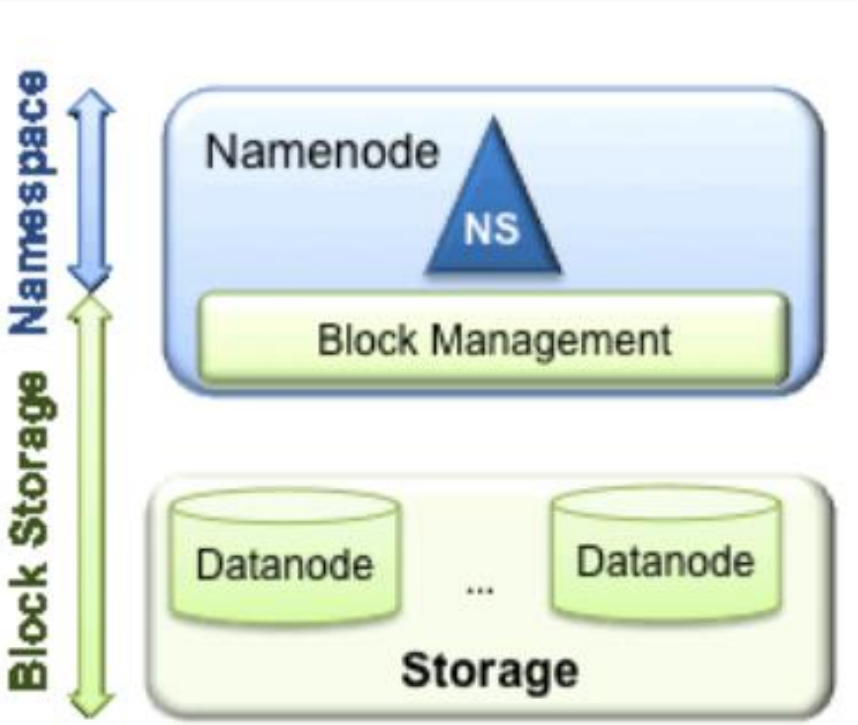
Arsitektur HDFS



HeartBeat pada HDFS



HDFS



HDFS mempunyai 2 layer utama:

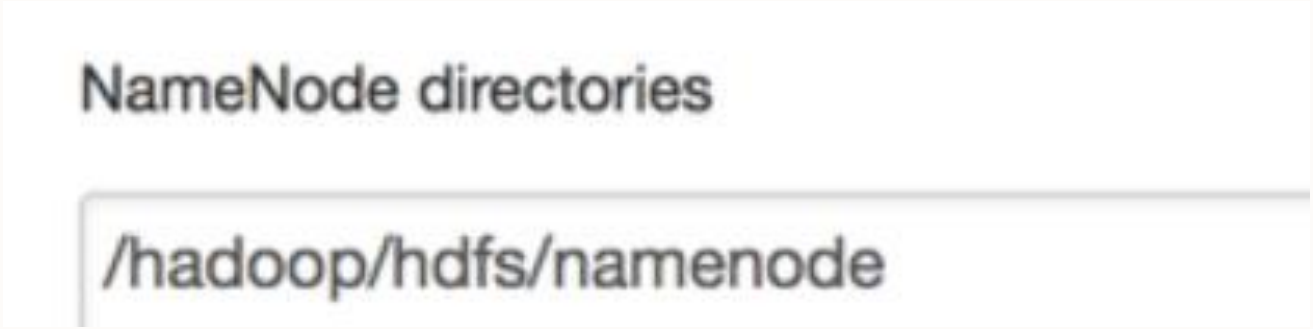
- **Namespace**
  - Terdiri dari directory, file, dan block.
  - Mendukung semua operasi sistem file terkait namespace seperti *create*, *delete*, *modify*, dan *list* file dan directory.
  - Namespace = lokasi tertentu di HDFS
- **Block Storage Service**, yang memiliki dua bagian: Namenode dan Datanode

Namenode dan Datanodes

- Master/slave architecture.
- HDFS terdiri dari sebuah **Namenode**, sebuah master server yang melakukan manajemen file seperti file sistem Namespace (file atau directory) dan mengatur akses file-file oleh client.
- Terdiri dari beberapa **DataNodes**.
- DataNodes melakukan menyimpan data dan pemrosesannya.
- File data terbagi kedalam satu atau beberapa blok dan satu set blok tersimpan dalam DataNodes.
- DataNodes melayani *read, write*, menjalankan *block creation, deletion*, dan *replication* berdasarkan instruksi dari Namenode.

File system Metadata

- Namespace disimpan dalam Namenode.
- Namenode menggunakan transaction log bernama **EditLog** untuk menyimpan setiap perubahan yang terjadi terhadap data di dalam metadata. Contohnya, membuat file baru atau perubahan replication factor.
- Keseluruhan file sistem Namespace, termasuk pemetaan blok-blok suatu file data dan propertinya, disimpan dalam suatu file bernama **FsImage**.



Distributed File Systems

Comparison of distributed file systems

Add languages

Article Talk | Read Edit View history

From Wikipedia, the free encyclopedia

See also: Comparison of distributed parallel fault-tolerant file systems

In computing, a distributed file system (DFS) or network file system is any file system that allows access to files from multiple hosts sharing via a computer network. This makes it possible for multiple users on multiple machines to share files and storage resources.

Distributed file systems differ in their performance, mutability of content, handling of concurrent writes, handling of permanent or temporary loss of nodes or storage, and their policy of storing content.

Locally managed [edit]

FOSS [edit]

Client	Written in	License	Access API	High availability	Shards	Efficient Redundancy	Redundancy Granularity	Initial release year	Memory requirements (GB)
Alluxio (Virtual Distributed File System)	Java	Apache License 2.0	HDFS, FUSE, HTTP/REST, S3	hot standby	No	Replication[1]	File[2]	2013	
Ceph	C++	LGPL	librados (C, C++, Python, Ruby), S3, Swift, FUSE	Yes	Yes	Pluggable erasure codes[3]	Pool[4]	2010	1 per TB of storage
Coda	C	GPL	C	Yes	Yes	Replication	Volume[5]	1987	
GlusterFS	C	GPLv3	libglusterfs, FUSE, NFS, SMB, Swift,	mirror	Yes	Reed-Solomon[6]	Volume[7]	2005	

Sumber: https://en.wikipedia.org/wiki/Comparison_of_distributed_file_systems

Komponen Hadoop – MapReduce

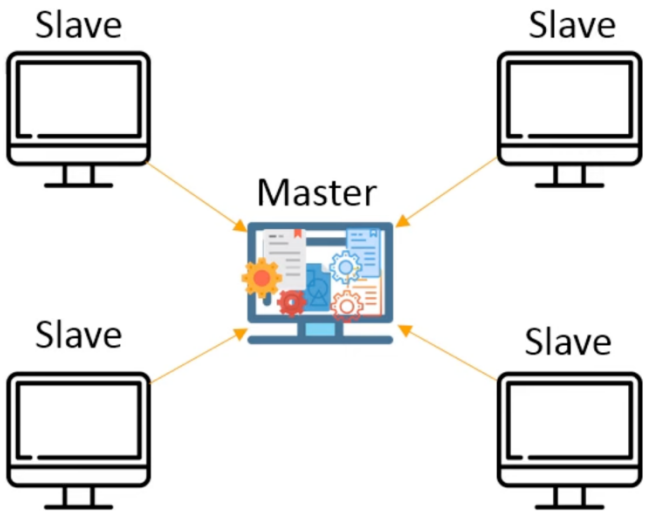
Hadoop MapReduce is a programming technique where huge data is processed in a parallel and distributed fashion



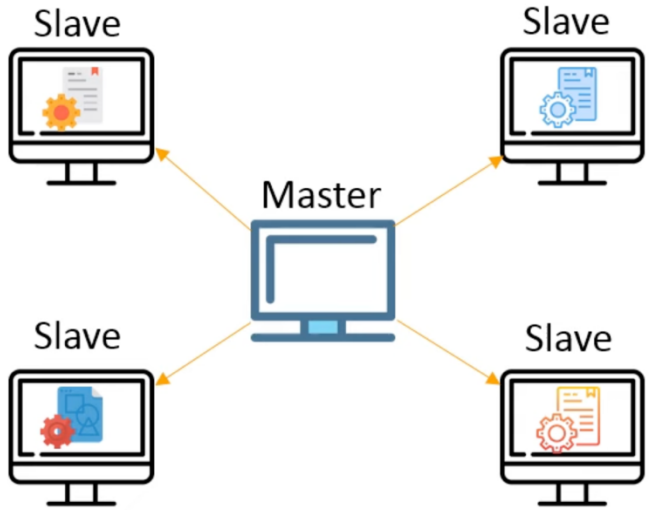
MapReduce is used for parallel processing of the Big Data, which is stored in HDFS

Pendekatan MapReduce

In MapReduce approach, processing is done at the slave nodes and the final result is sent to the master node

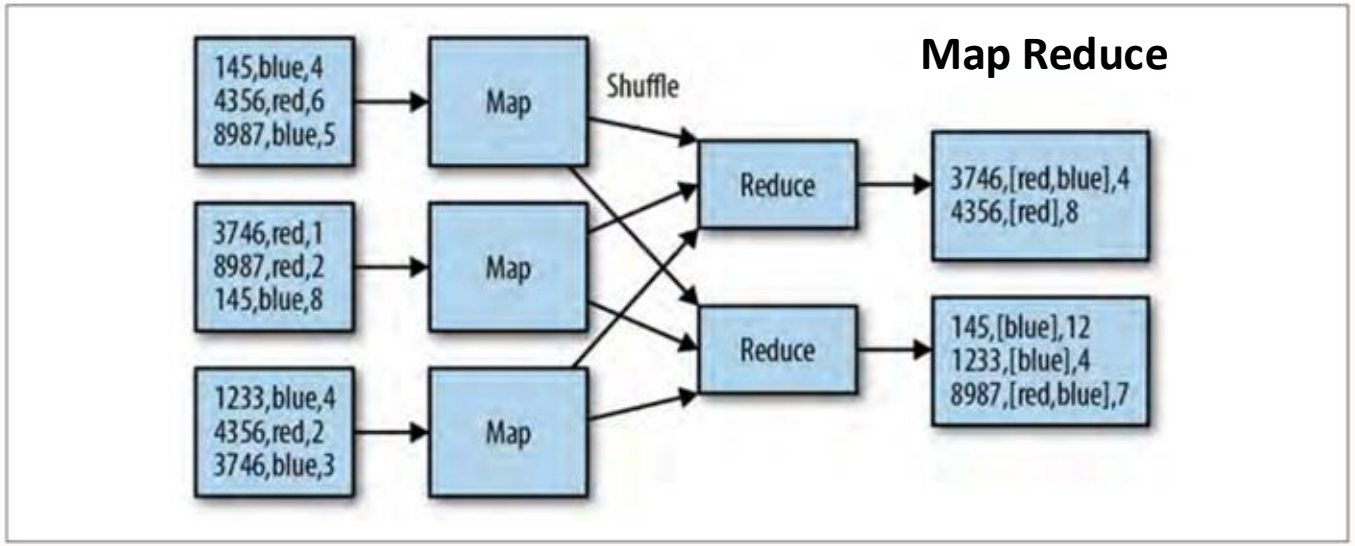


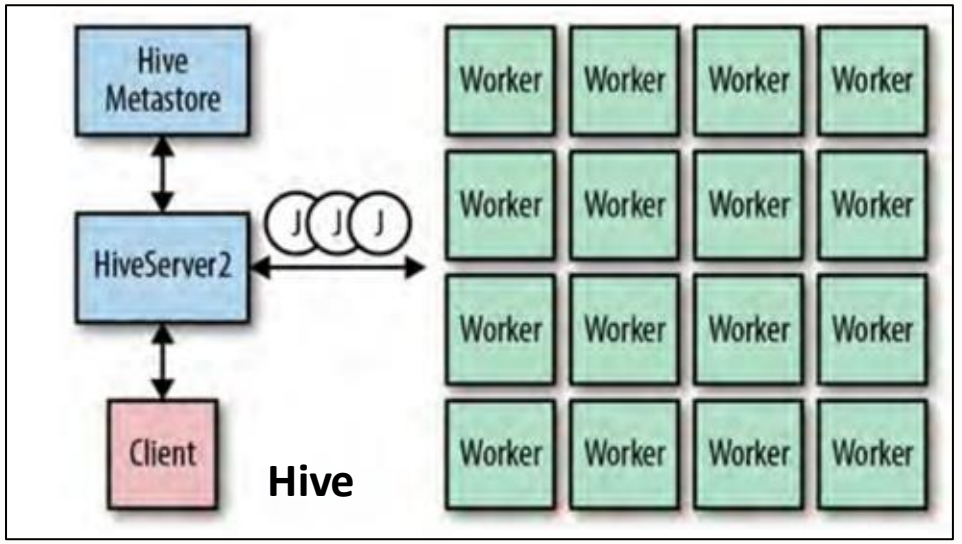
Traditional approach – Data is processed at the Master node

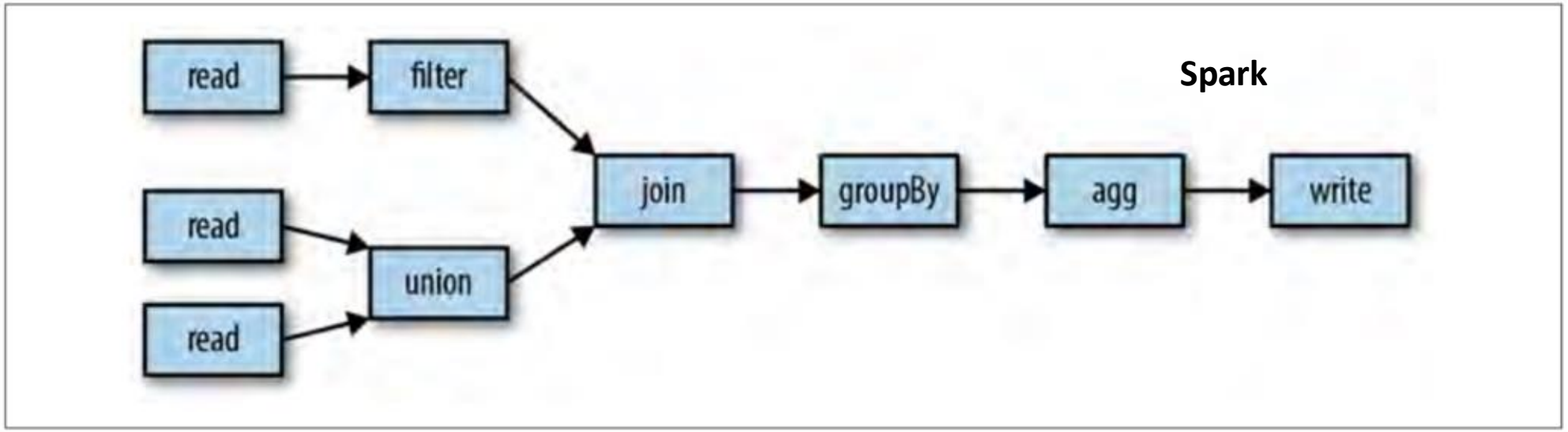


MapReduce approach – Data is processed at the Slave nodes

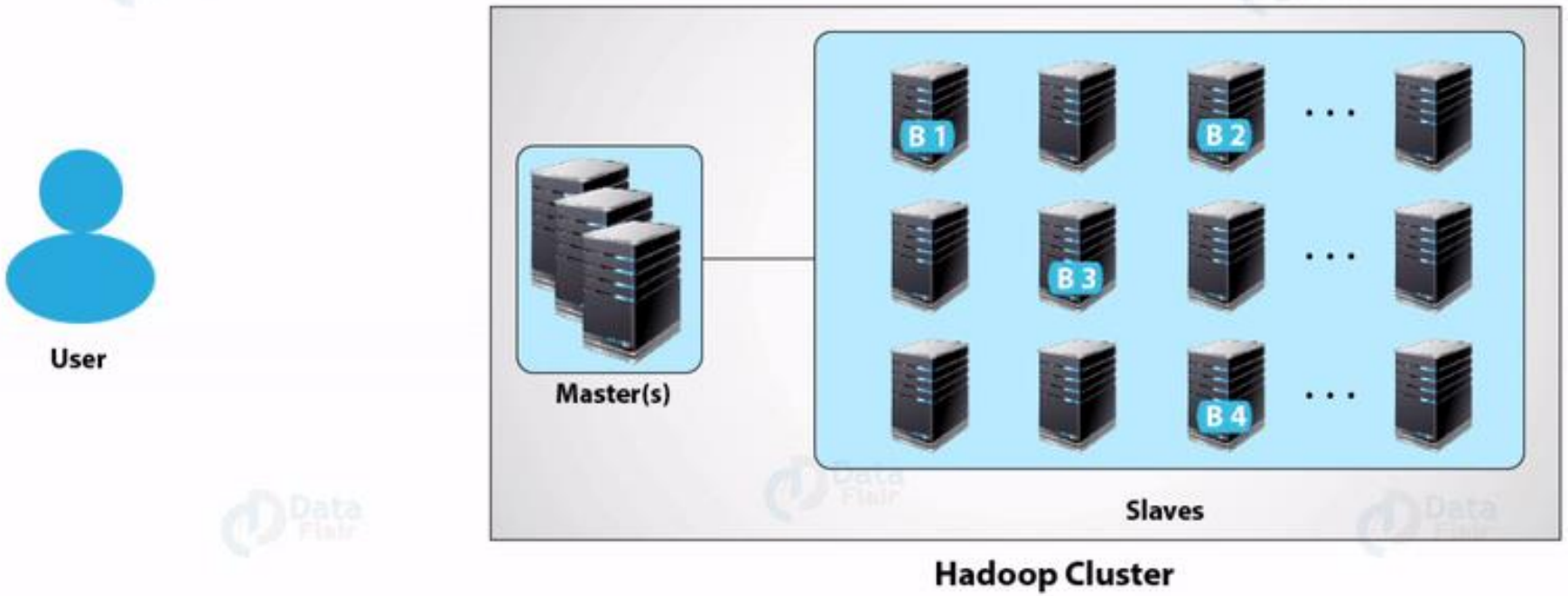
Komponen Hadoop – MapReduce





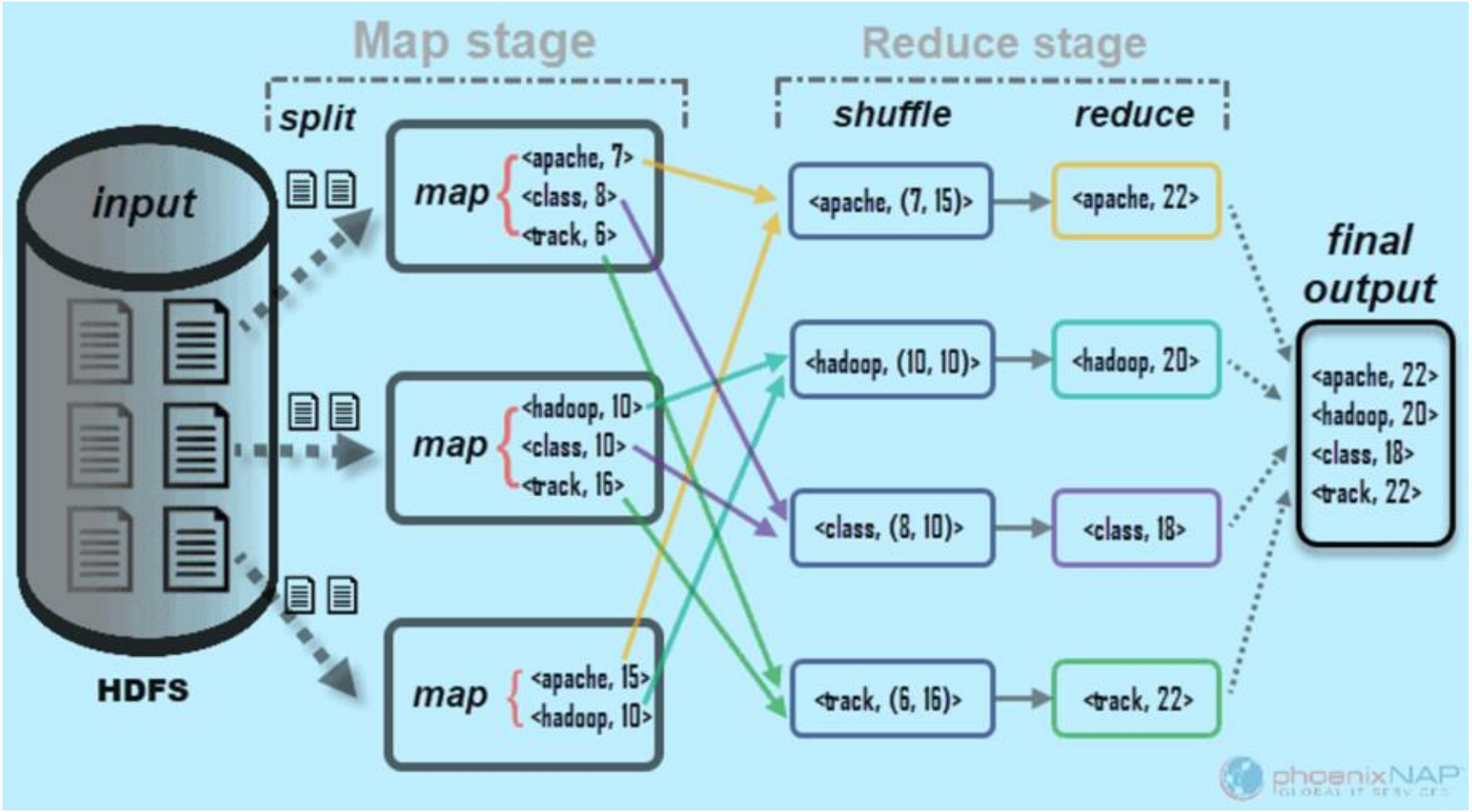


How MapReduce works



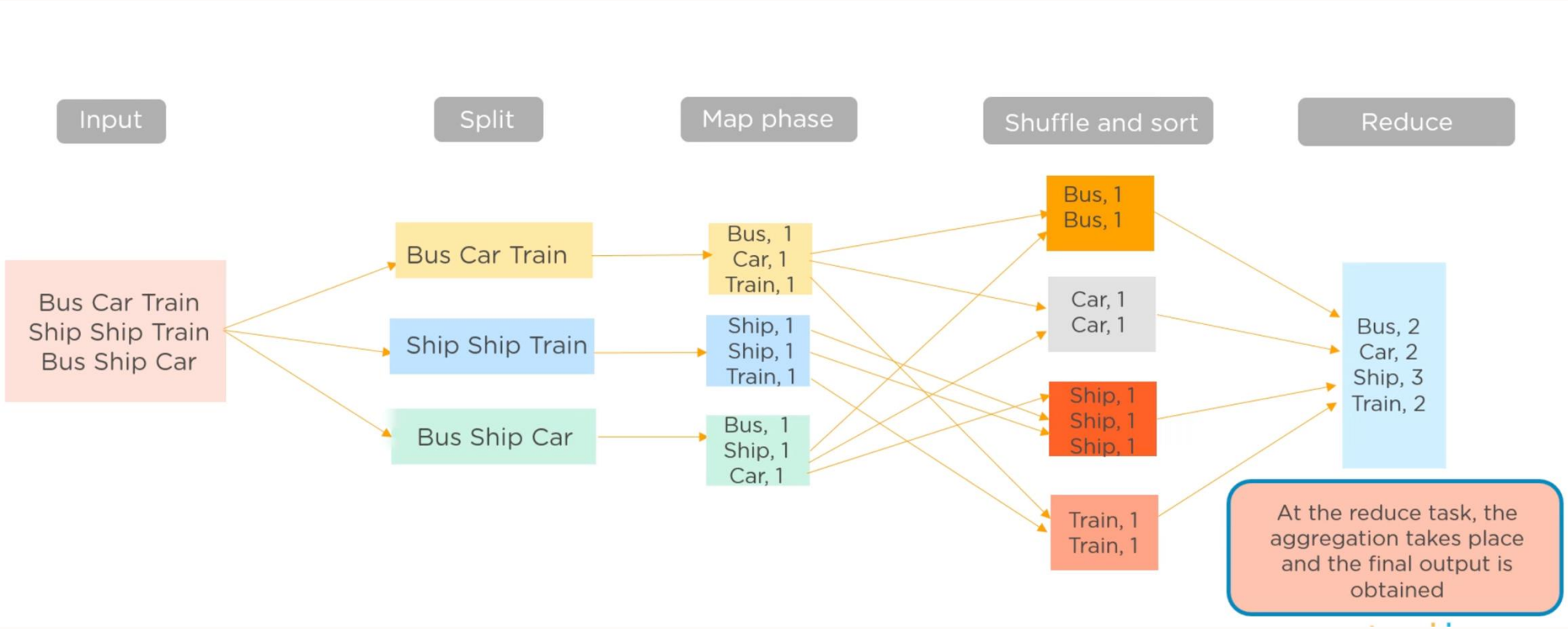
Instead of moving data to the processing unit, we are moving the processing unit to the data in the MapReduce Framework

Contoh MapReduce menggunakan 3 nodes untuk job WordCount()

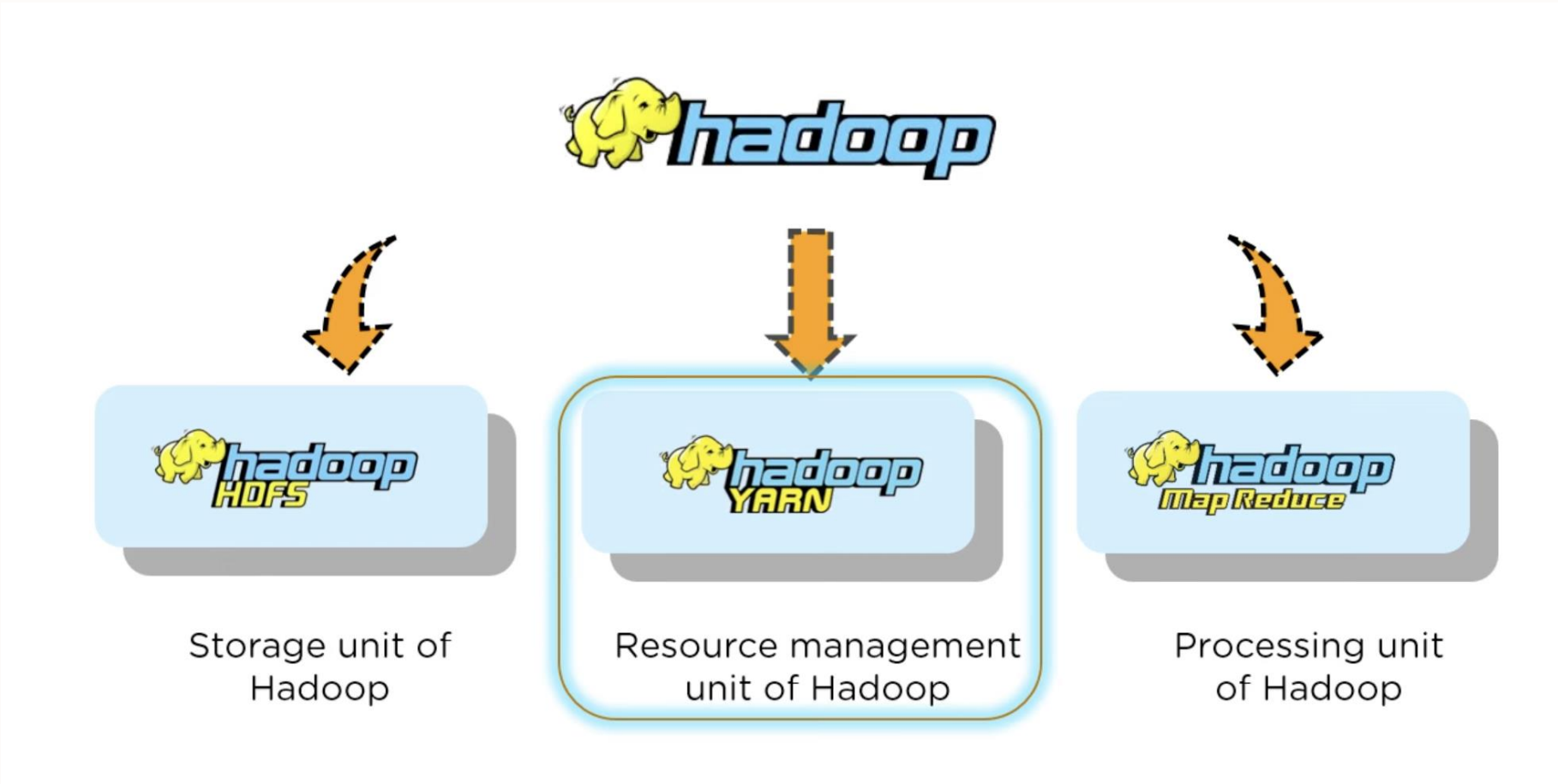


key-value pairs

Contoh MapReduce



Hadoop Versi >= 2.0

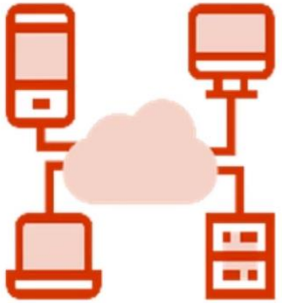


YARN (Yeat Another Resource Negotiator)

YARN – Yet Another Resource Negotiator



Acts like an OS to Hadoop 2

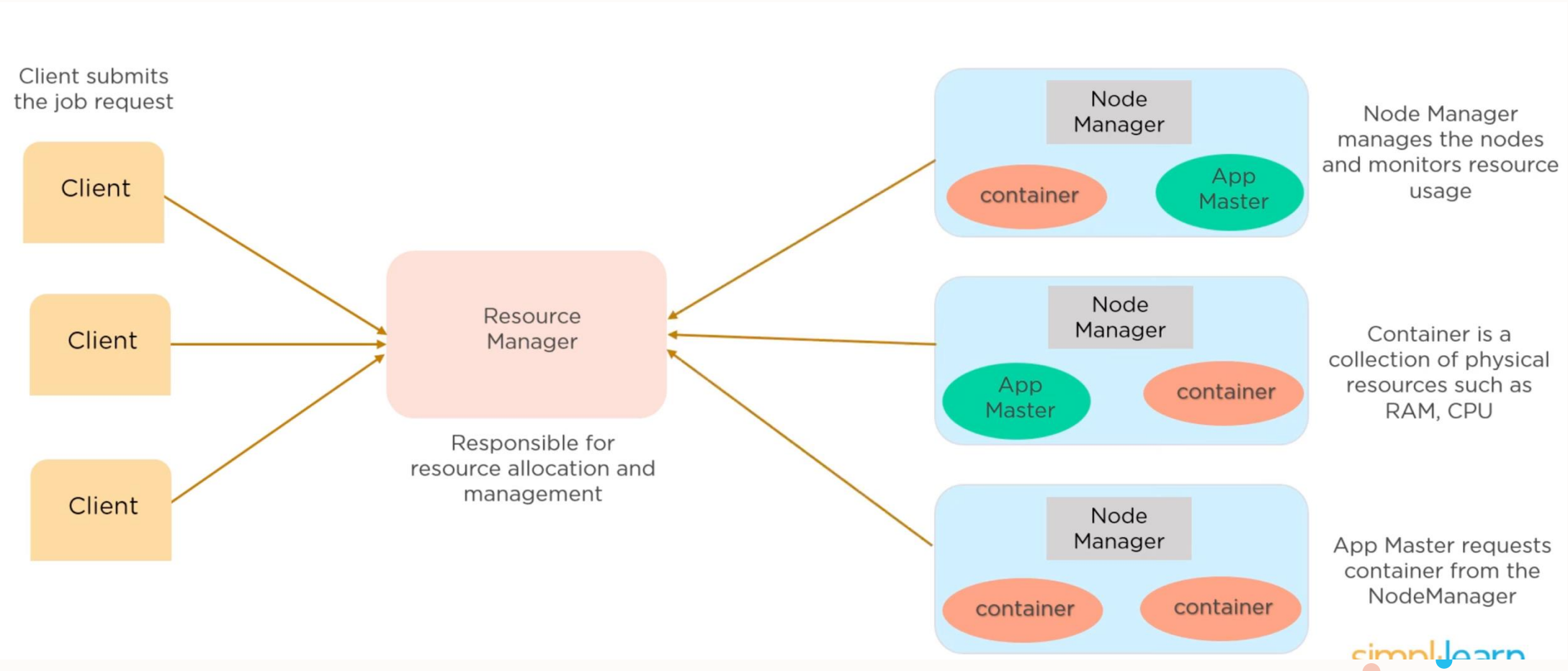


Responsible for managing cluster resources



Does job scheduling

YARN


Client submits the job request
Client
Client
Client
Resource Manager
Responsible for resource allocation and management
Node Manager
container
App Master
Node Manager manages the nodes and monitors resource usage
Node Manager
App Master
container
Container is a collection of physical resources such as RAM, CPU
Node Manager
container
container
App Master requests container from the NodeManager

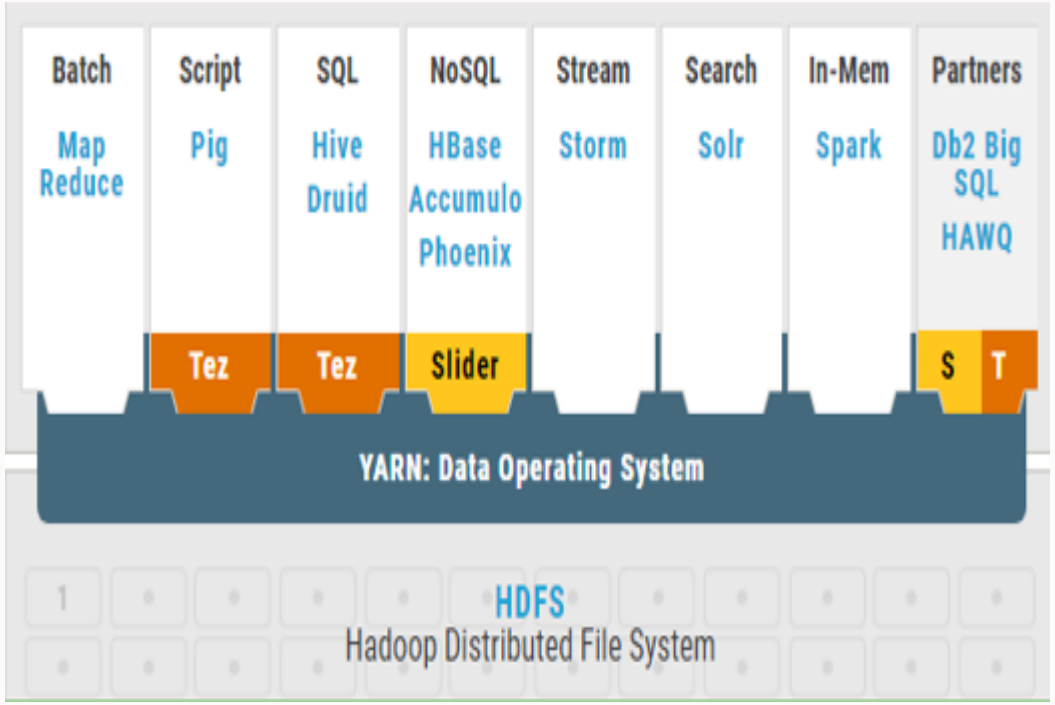
Komponen Hadoop - Yarn

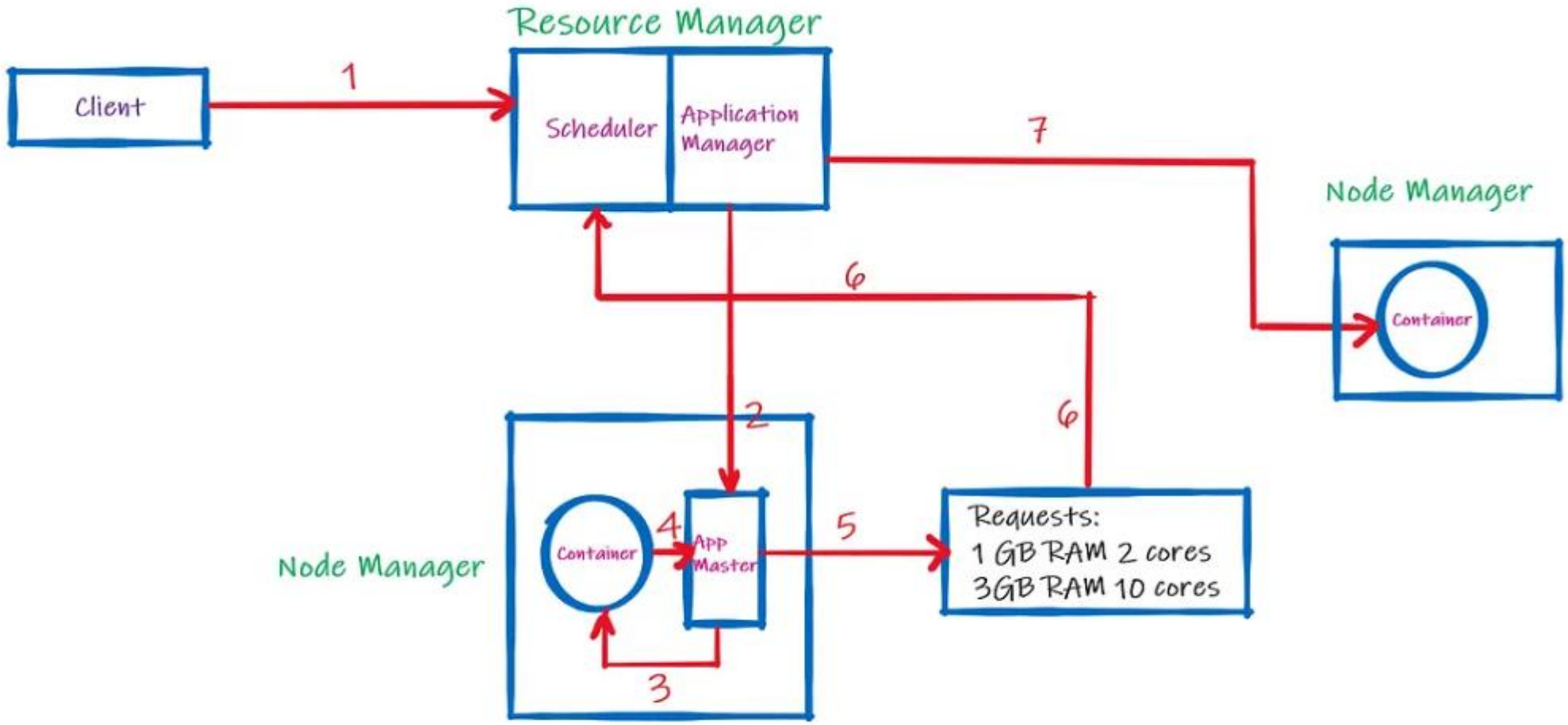
YARN memungkinkan berbagai data processing engines:

- interactive processing,
- graph processing,
- batch processing,
- stream processing

untuk menjalankan dan memproses data yang disimpan dalam HDFS (*lihat gambar arsitektur Hadoop/HDP*)

- **Resource Manager**: Scheduler (pengalokasian resource e.g. CPU, Memory) and Application Manager
- **Node Manager**

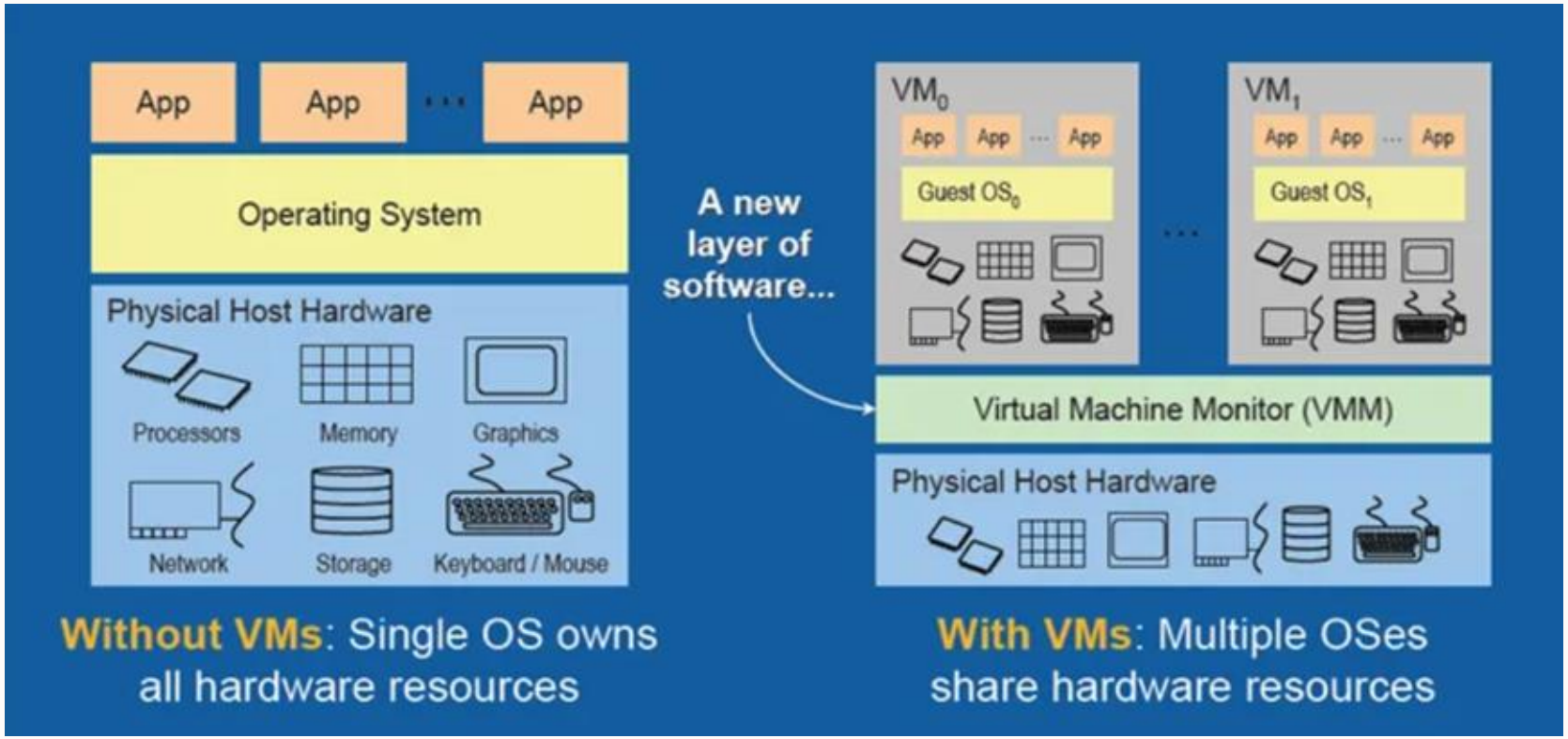




YARN Architecture

Pengenalan Virtualization

- **Virtualization technology** adalah teknologi pembuatan sesuatu versi virtual – bukan aktual -, seperti sistem operasi (OS), server, storage, atau network.



Implementasi Virtualisasi

- Server Virtualization
- Desktop Virtualization
- Storage Virtualization
- Virtualization vs. Cloud Computing *complementary, not interchangeable*





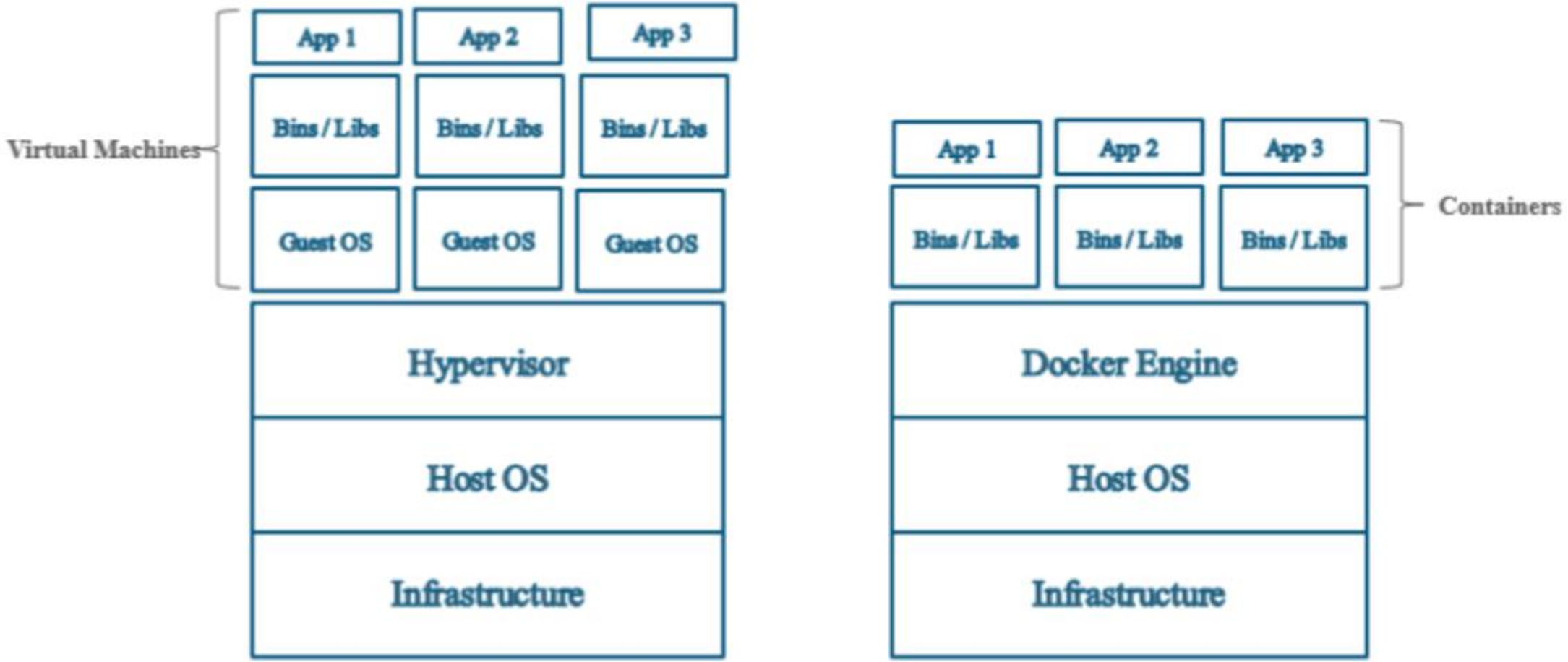








Virtual Machine vs Container

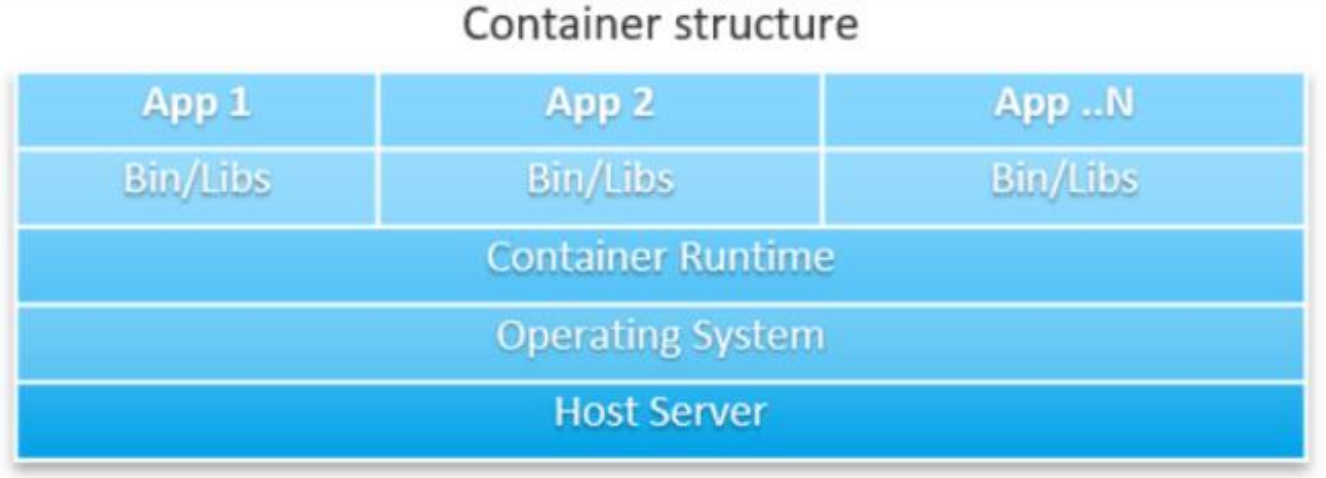


Containerization

Container adalah sebuah virtualisasi OS yang dapat membungkus suatu aplikasi beserta dependency dan environment-nya. Setiap container memiliki process yang terisolir sehingga tidak mengganggu host OS ataupun container yang lain.

Keuntungan Container:

- Portability
- Resource density and efficiency
- Isolation and resource sharing
- Speed
- Scalability



Praktikum: Hadoop

- Hortonworks Data Platform (HDP) Sandbox
- VirtualBox

Perintah Shell

- Hadoop menyertakan berbagai perintah mirip shell yang berinteraksi langsung dengan HDFS dan sistem file lain yang didukung Hadoop.
- Perintah **`hdfs dfs -help`** mencantumkan perintah yang didukung oleh Hadoop shell.
- Selanjutnya, perintah **`hdfs dfs -help command-name`** menampilkan bantuan yang lebih detail untuk sebuah perintah.
- Perintah-perintah ini mendukung sebagian besar operasi sistem file normal seperti menyalin file, mengubah hak akses file, dll. Selain itu, juga mendukung beberapa operasi khusus HDFS seperti mengubah replikasi file.

HDFS Configuration

HDFS Defaults

- Block Size – 64 MB
- Replication Factor – 3
- Web UI Port – 50070

HDFS conf file - `/etc/hadoop/conf/hdfs-site.xml`

```
<property>
    <name>dfs.namenode.name.dir</name>

          <value>file:///data1/cloudera/dfs/nn,file:///data2/cloudera/dfs/nn</va
lue>
 </property>

 <property>
    <name>dfs.blocksize</name>
    <value>268435456</value>
 </property>

 <property>
    <name>dfs.replication</name>
    <value>3</value>
 </property>

 <property>
    <name>dfs.namenode.http-address</name>
    <value>itracXXX.cern.ch:50070</value>
 </property>
```

Terima kasih

Thanks to:

- Apache Hadoop
- SimpliLearn (some images credited to)